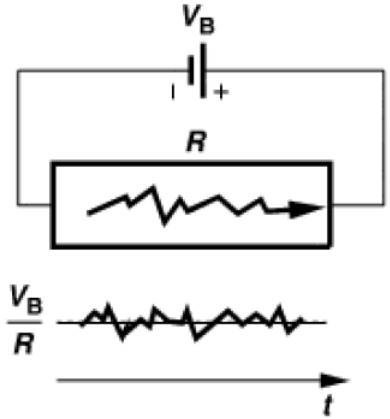


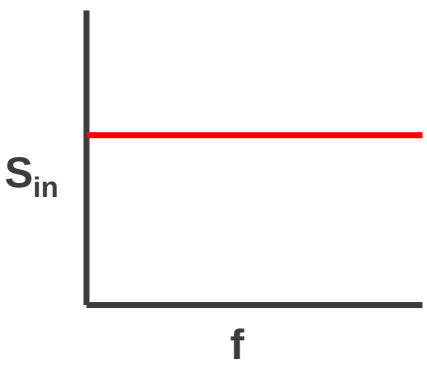
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BRUSSEL

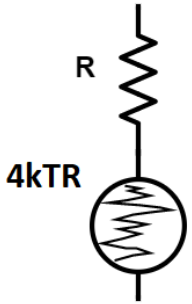
Notions of Noise

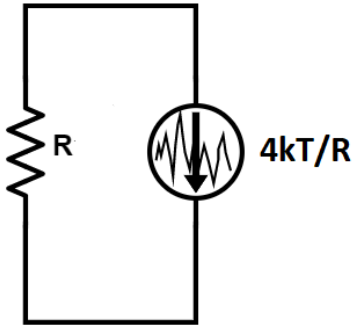
Arno, Chang-Shao, Ezgi, Xuepu

Resistor Noise







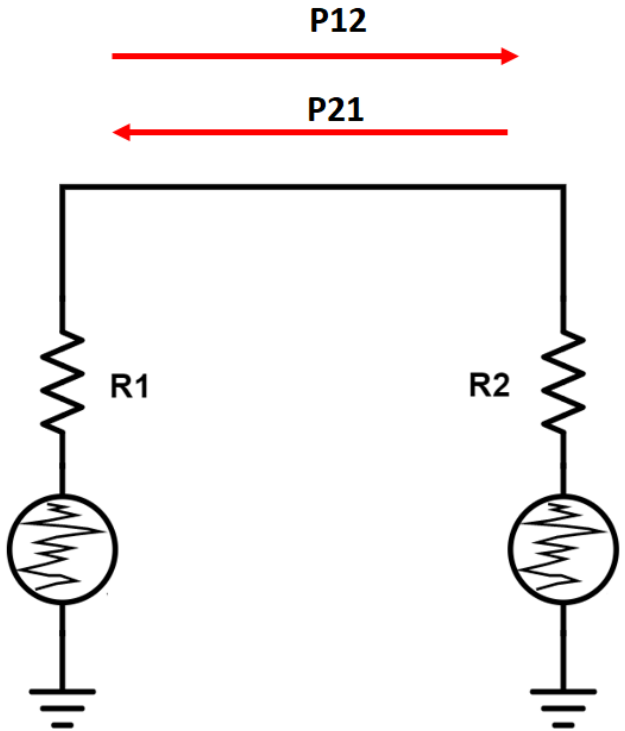


- Current in resistor connected to battery fluctuates because of thermal fluctuations.
- Modelled as equivalent series voltage or parallel current source.

Session 4: Noise

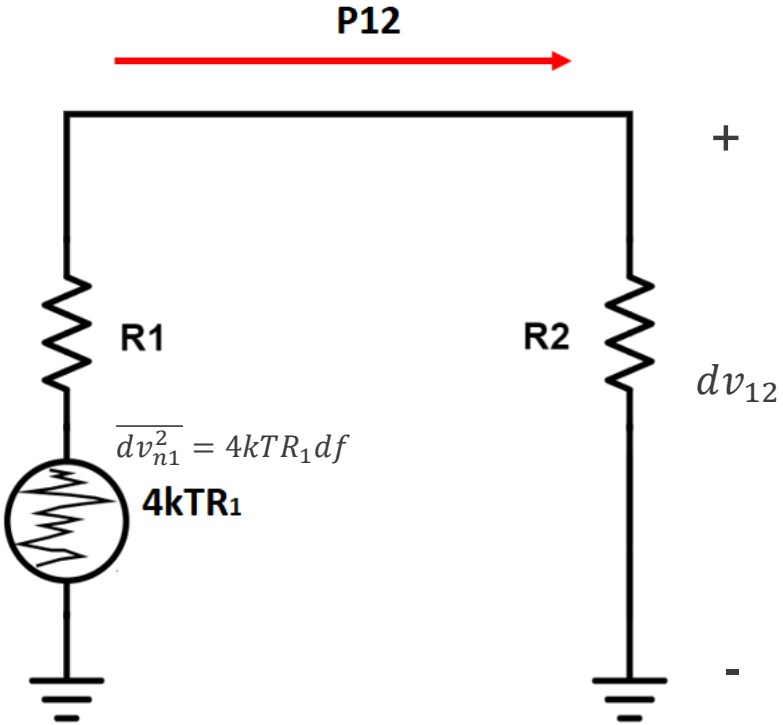
- Solve exercise 1
  - ETA: 10 min

Exercise 1a



- Resistor R_1 and R_2 are connected in series.
- P_{12} is the power transferred from resistor R_1 to R_2 due to thermal noise in resistor R_1 in an arbitrary bandwidth B. Similarly P_{21} is the power transferred from resistor R_2 to R_1.
- Calculate P_{12} and P_{21} ?
- What is the relationship between P_{12} and P_{21} Can you comment on this?

Solution 1a



By voltage divider,

$$\overline{dv_{12}^2} = 4kTR_1\left(\frac{R_2}{R_1+R_2}\right)^2 df$$

$$\overline{P_{12}} = \int_0^B \frac{\overline{dv_{12}^2}}{R_2} df = 4kBT\frac{R_1R_2}{(R_1+R_2)^2}$$

Similarly,

$$\overline{P_{21}} = \int_0^B \frac{\overline{dv_{21}^2}}{R_1} df = 4kBT\frac{R_1R_2}{(R_1+R_2)^2}$$

$\overline{P_{12}} = \overline{P_{21}}$ is expected since it is thermal equilibrium

Maximum power transfer condition: $R_2 = R_1$ $(Z_2 = Z_1^*)$

$$\overline{P_{12}} = \overline{P_{21}} = kBT$$

Some textbook just estimates the noise power from a resistor to be ***kBT***
(since it is the worst case)

Exercise 1b-c

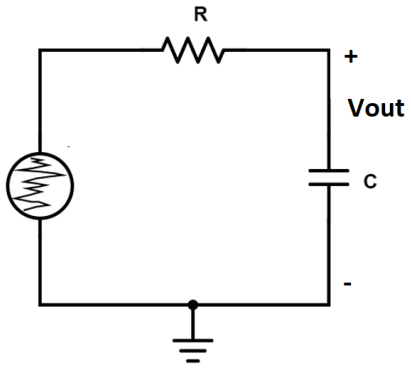


Figure A

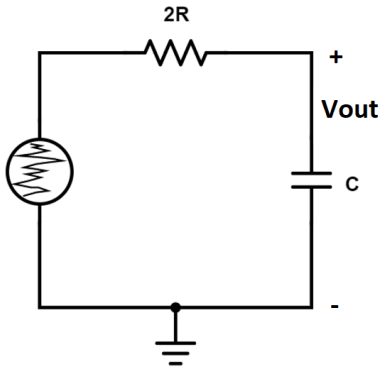


Figure B

- Resistor R and capacitor C are connected in series.
- In Figure A, calculate integrated noise across the capacitor? Also calculate the same when resistor R is increased to 2R as in Figure B?
- Plot the noise voltage power spectral density in both cases across the capacitors with frequency? Take R=1kΩ and C=1pF , T=300 K , k=$1.38 * 10^{-23}$ JK^{-1}. In the second case change the value of the resistance to 2kΩ.

Solution 1b-c

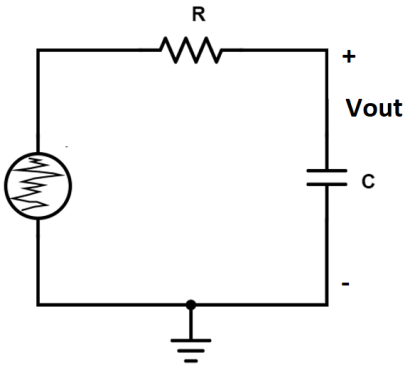


Figure A

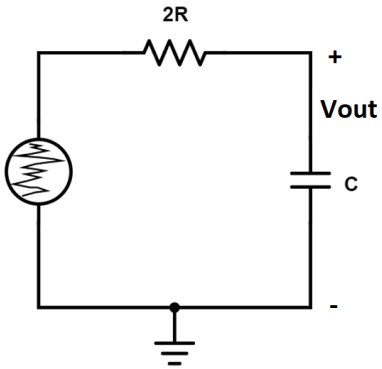


Figure B

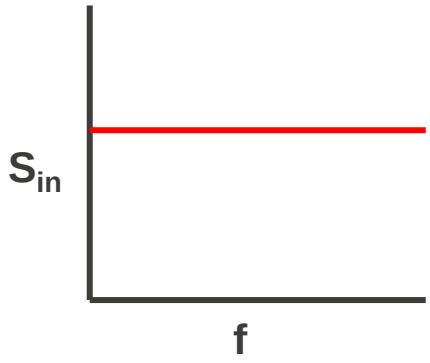
By voltage divider, $(s = j\omega = j2\pi f)$

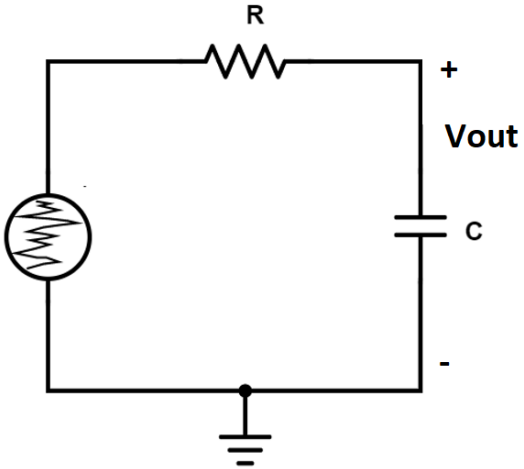
$$\overline{dv_{n,out}^2} = 4kTR\left|\frac{1/(sC)}{R+1/(sC)}\right|^2 df = 4kTR\frac{1}{1+(2\pi fCR)^2}df$$

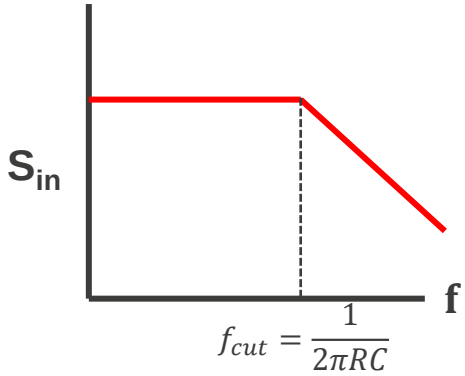
Integrated noise $\overline{v_{n,out}^2} = \int_0^\infty \overline{dv_{n,out}^2}\, df = \frac{kT}{C}$

$\overline{v_{out}^2}$ is independent of R, why?

Solution 1b-c





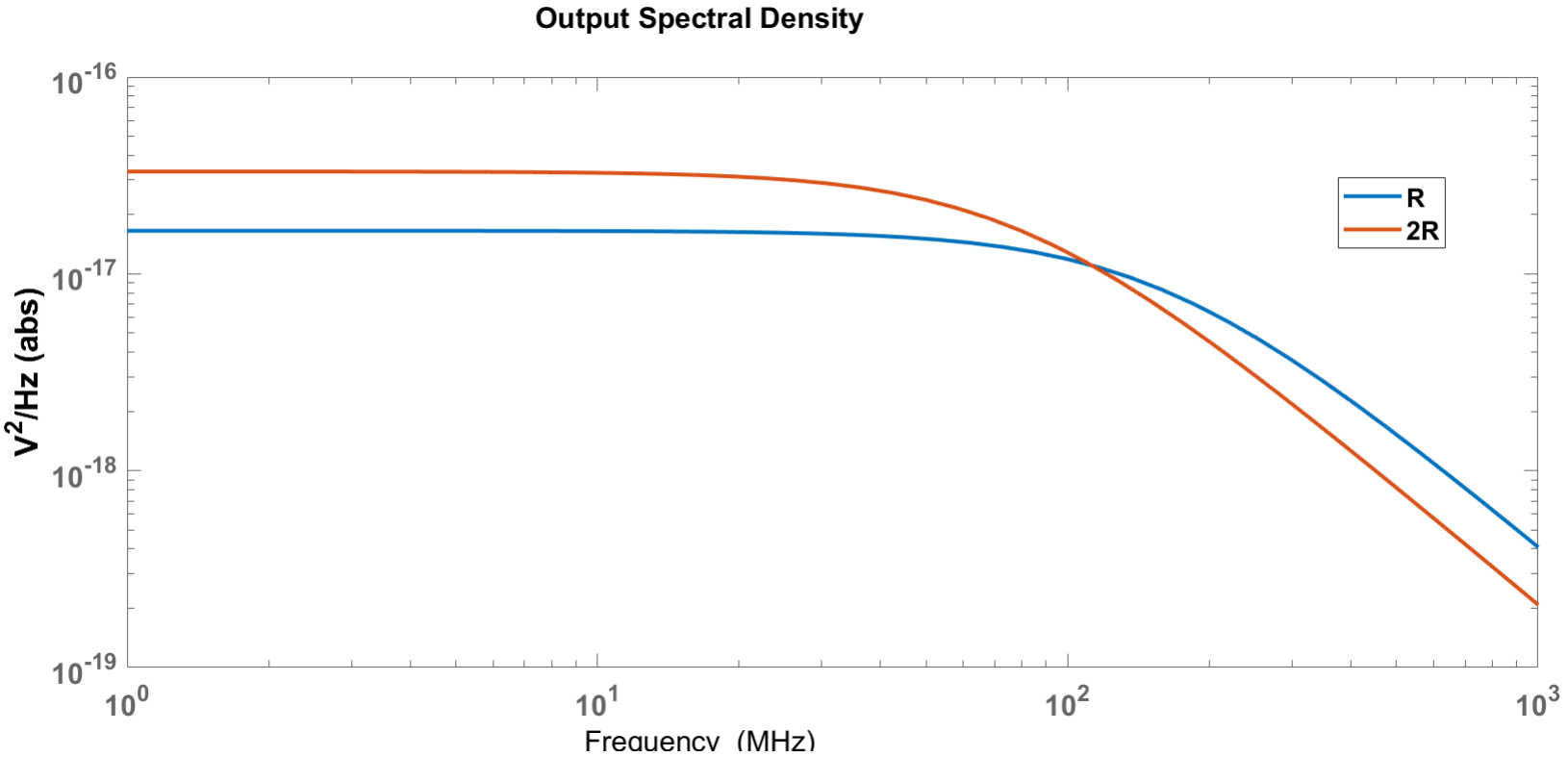


- $\overline{dv_{n,out}^2} = \overline{dv_{n,in}^2}\,|TF(f)|^2$
- $\overline{dv_{n,out}^2} = (4kTR)\frac{1}{1+(2\pi fCR)^2}\,df$
- $\overline{v_{n,out}^2} = \int_0^\infty \overline{dv_{n,out}^2}\,df$
- $\overline{v_{n,out}^2} = \frac{kT}{C}$
- Changing R to 2R does not change integrated noise.

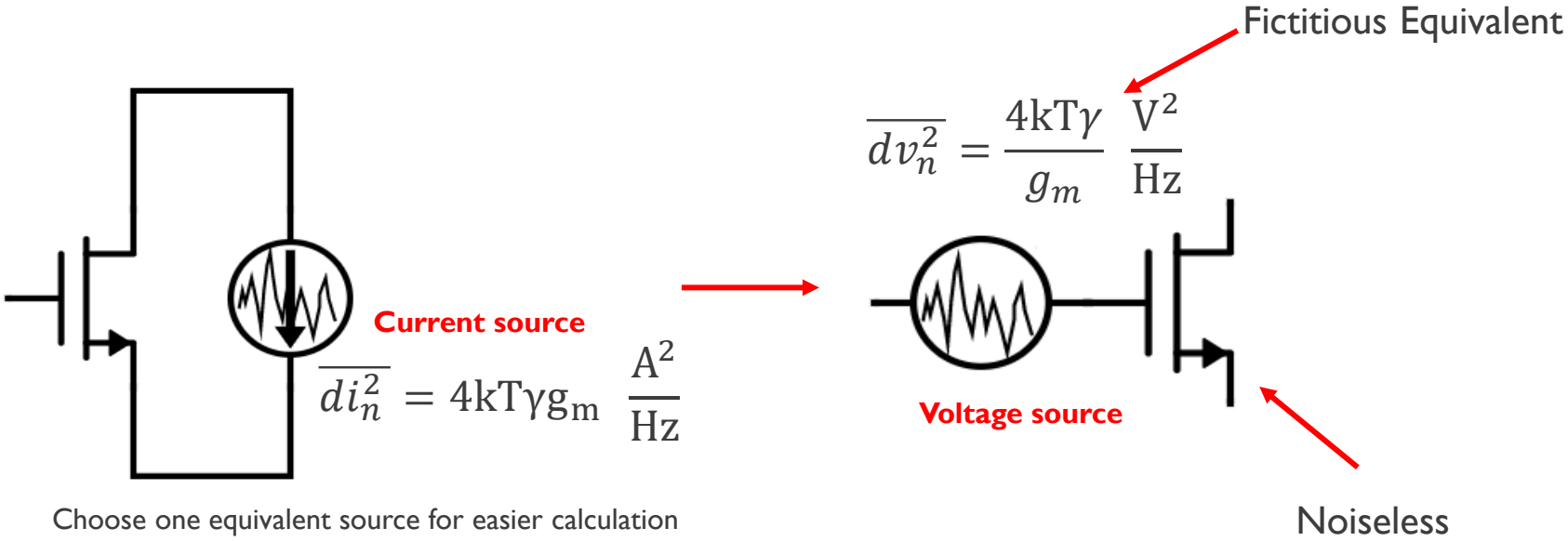
As R ↑, the noise power from R ↑, but the cutoff frequency f_{cut} ↓, meaning more noise power gets attenuated
Changing C changes f_{cut}, while the noise power from R is constant

Refer to 4.3.2 in the lecture note

Solution 1b-c



Thermal Noise/Flicker noise in MOS transistor



Choose one equivalent source for easier calculation

- Linear superposition rules still applies
  - When calculating for one source, short other voltage sources and open other current sources
  - Total noise is the sum of contribution from every source

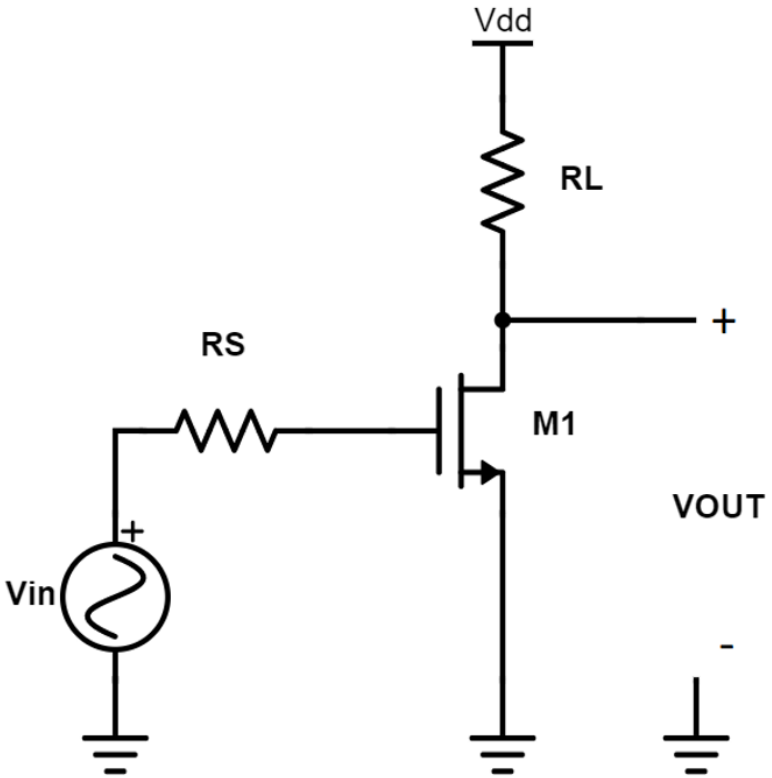
Flicker noise:

$$\overline{di_d^2} = \frac{g_m^2 K_f(V_{GS}, V_{DS}, V_{SB})}{C'_{ox}} \frac{1}{WL} \frac{1}{f} \Delta f$$

Session 4: Noise

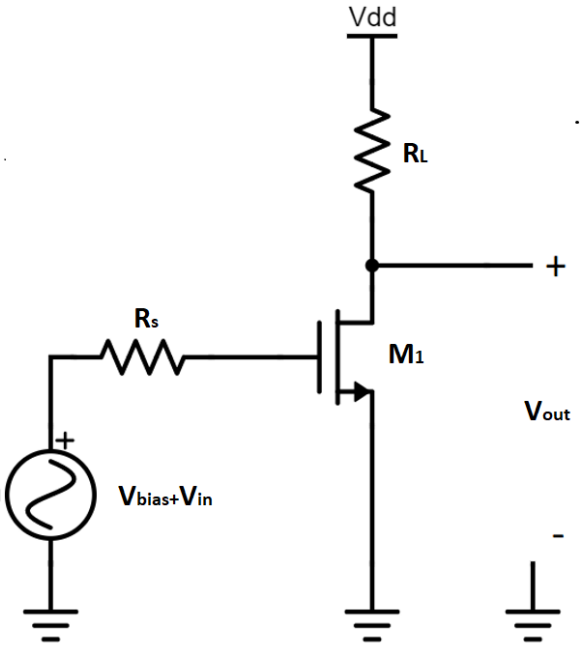
- Solve exercise 2
  - ETA: 10 min

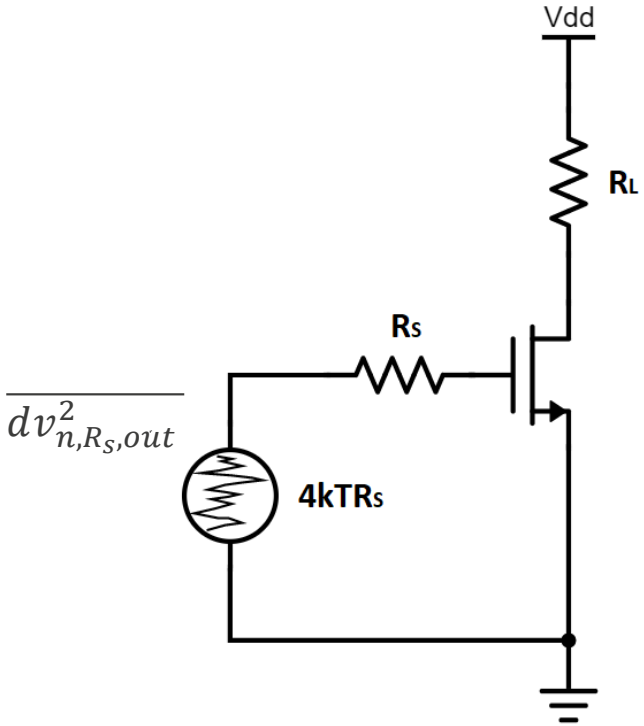
Exercise 2a



- Calculate output and input referred noise of common source amplifier?
- Assume transistor properly biased with infinite output impedance and trans-conductance gm.
- Consider only thermal noise for calculations.
- How will you decrease by a factor of 2 the input referred thermal noise contribution of M_1 and load resistor R_L keeping the gain of the amplifier same ?

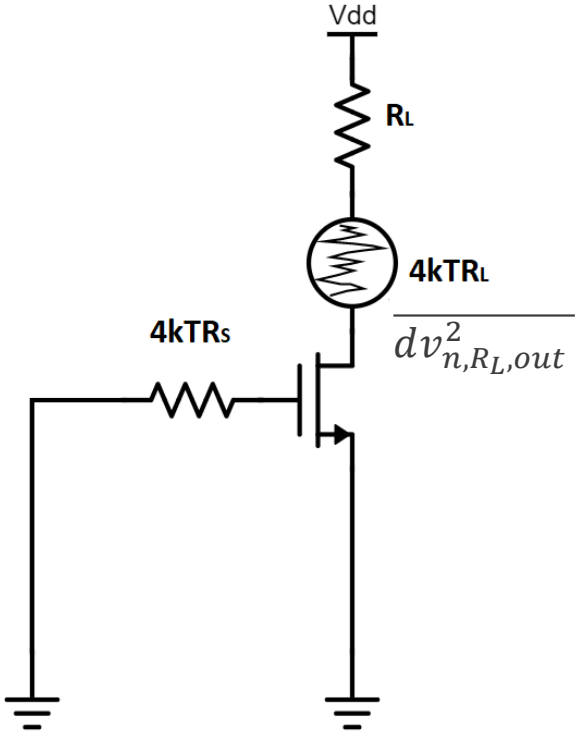
Solution 2a

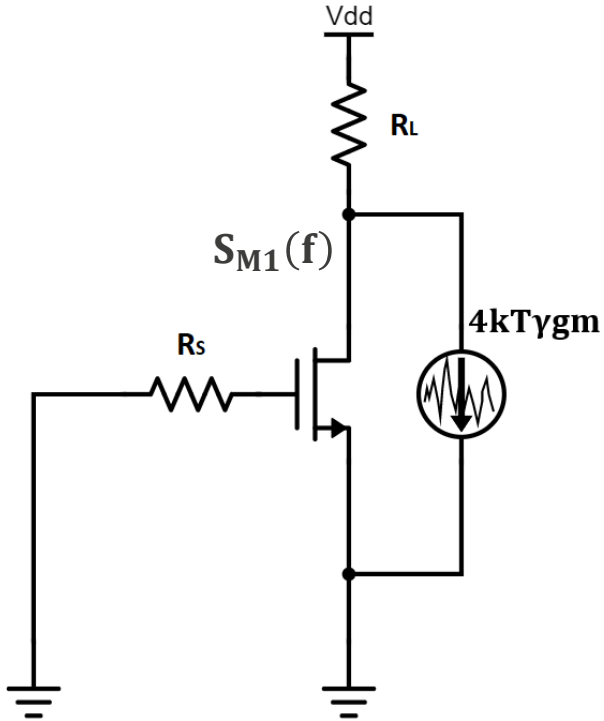




- $A_v = -g_m R_L$
- $\overline{dv^2_{n,R_S,out}} = 4kTR_s(g_m R_L)^2 df$

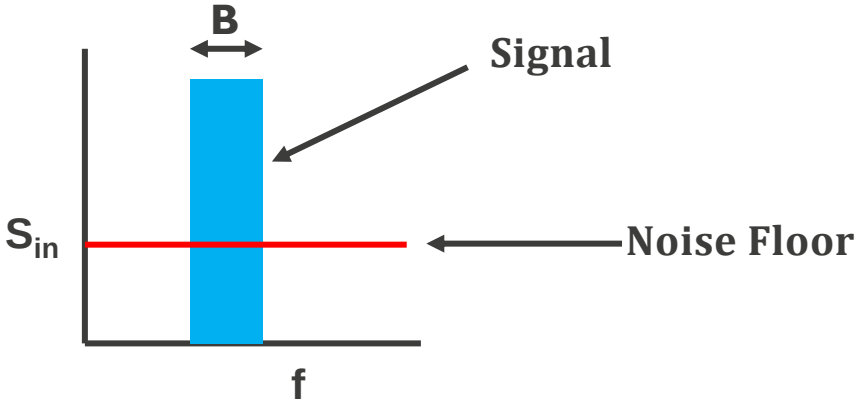
Solution 2a





- $\overline{dv^2_{n,R_L,out}} = 4kTR_L df$
- $\overline{dv^2_{n,M1,out}} = 4kT\gamma gmR_L^2 df$

Solution 2a

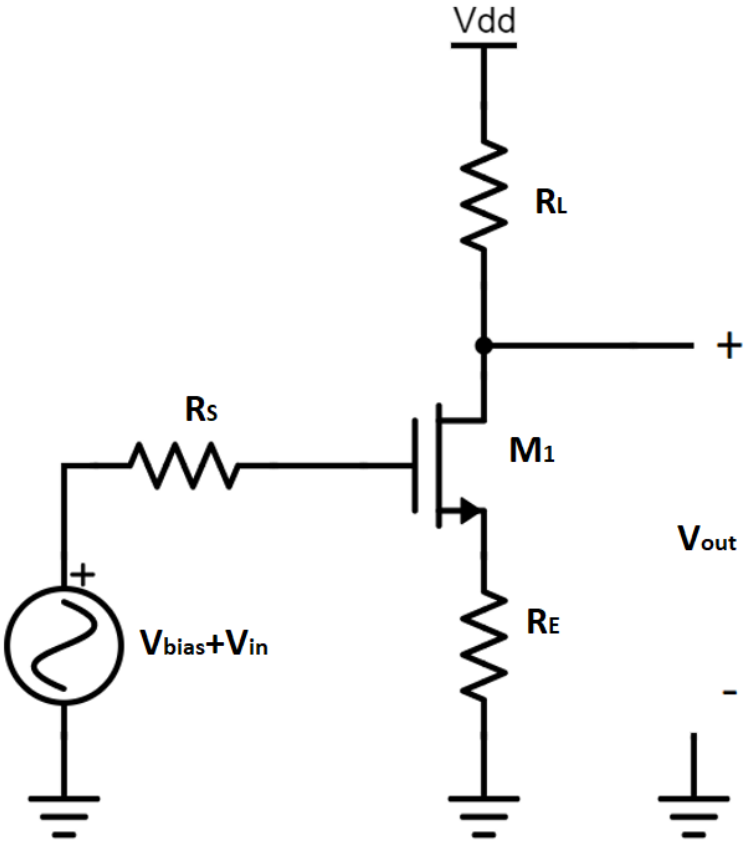


- $A_v = -g_m R_L$
- $\overline{dv_{n,out}^2} = \overline{dv_{n,R_S,out}^2} + \overline{dv_{n,M1,out}^2} + \overline{dv_{n,R_L,out}^2} = \left[4kTR_S(g_m R_L)^2 + \frac{4kT\gamma}{g_m}(g_m R_L)^2 + 4kTR_L\right] df$
- $\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,out}^2}}{A_v^2} = \left[4kTR_S + \frac{4kT\gamma}{g_m} + \frac{4kT}{g_m^2 R_L}\right] df$
- $SNR_{out} = \frac{P_{sig}}{P_{noise}} = \frac{V_{in}^2 A_v^2}{\overline{dv_{n,out}^2} * B} = \frac{V_{in}^2}{\frac{\overline{dv_{n,out}^2}}{A_v^2} * B} = \frac{V_{in}^2}{\overline{dv_{n,in,eq}^2} * B}$ (Here S_{out} and S_{in}) are output and input referred noise. B is the signal bandwidth

Solution 2a

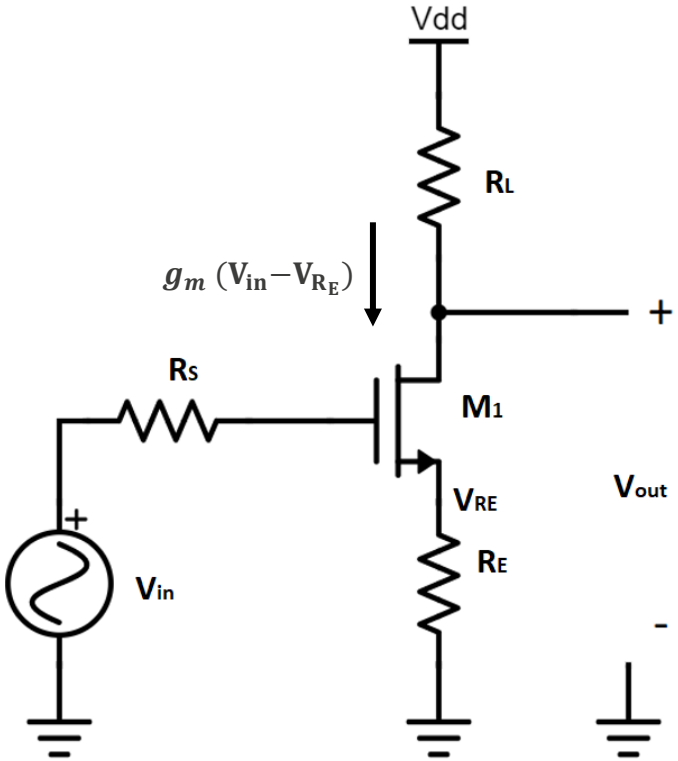
- $\overline{dv_{n,in,eq}^2} = 4\mathrm{kTR_s} + \frac{4\mathrm{kT}\gamma}{g_m} + \frac{4\mathrm{kTR_L}}{\mathrm{A_v^2}}$
- How to reduce the noise while keep the same gain?
- Decrease the $\mathrm{R_L}$ by half and increase gm by factor of 2 will keep the gain $\mathrm{A_v}$ same but will decrease input referred noise contribution by half of both transistor $\mathrm{M_1}$ and $\mathrm{R_L}$.
- Be aware of change of other metrics (poles/zeros, voltage swing, power consumption…)

Exercise 2b



- Calculate output and input referred noise of the common source degenerated amplifier in the figure?
- Assume transistor with infinite output impedance and trans-conductance gm.
- Is the input referred noise higher or lower compared common source amplifier? Consider only thermal noise for calculations.

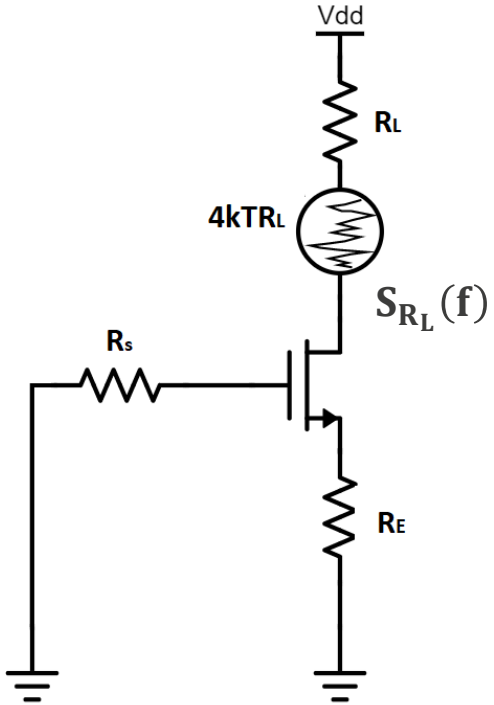
Solution 2b



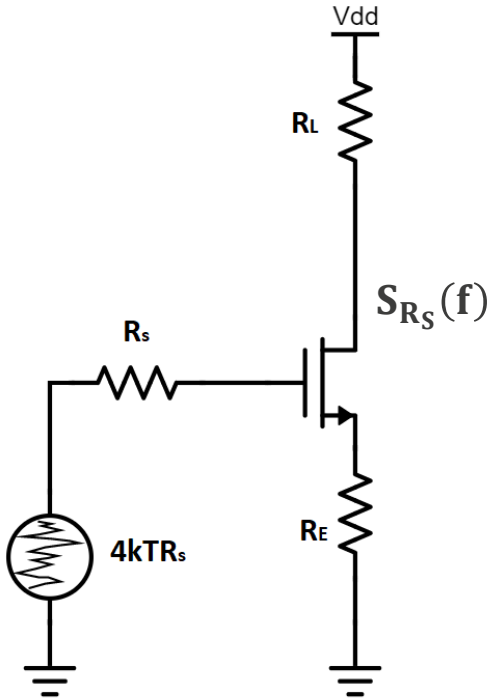
Gain calculation

- $g_m(V_{in}-V_{R_E}) = \frac{V_{R_E}}{R_E}$ thus $V_{R_E} = \frac{g_m R_E V_{in}}{1+g_m R_E}$
- $V_{out} = g_m(V_{in} - V_{R_E})R_L = \frac{-g_m R_L V_{in}}{1+g_m R_E}$
- $A_v = \frac{-g_m R_L}{1+g_m R_E}$

Solution 2b

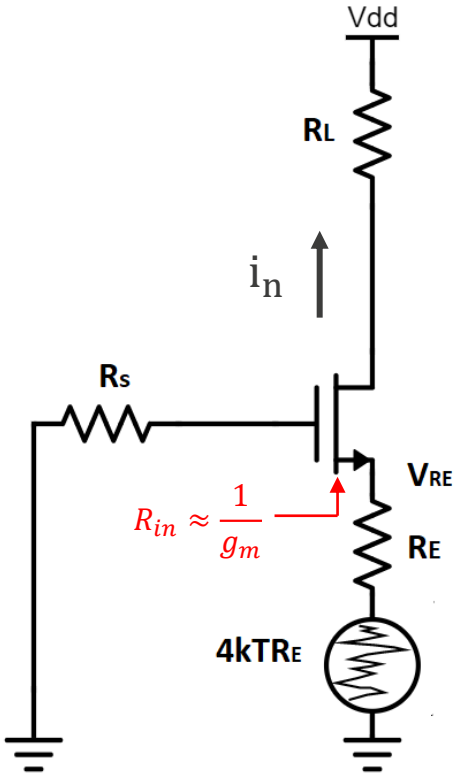


- $\overline{dv_{n,R_L,out}^2} = 4\mathrm{kTR_L}df$



- $\overline{dv_{n,R_s,out}^2} = 4\mathrm{kTR_s}\ (\mathrm{A_v})^2 df$

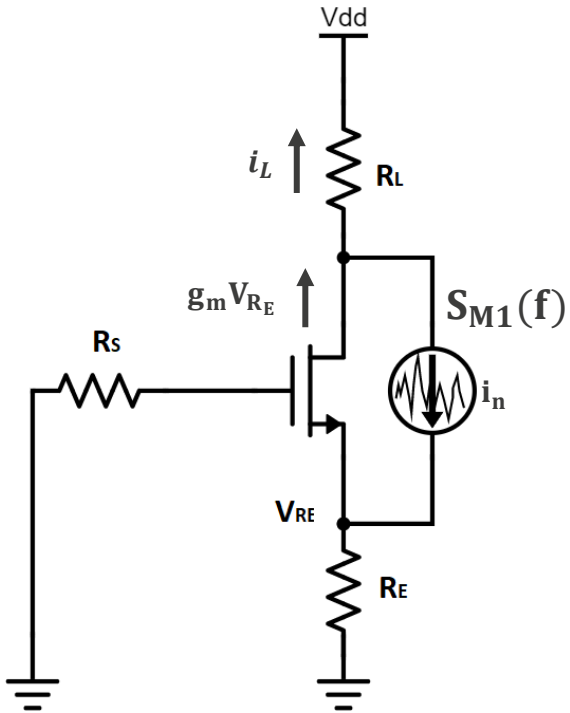
Solution 2b



- $\overline{i_n^2} = 4kTR_E \frac{1}{\left(\frac{1}{gm}+R_E\right)^2}$
- $\overline{dv_{n,R_E,out}^2} = \overline{i_n^2} R_L^2 = \frac{4kTR_E g_m^2 R_L^2}{(1+g_m R_E)^2}$

Refer to 5.8 in lecture note

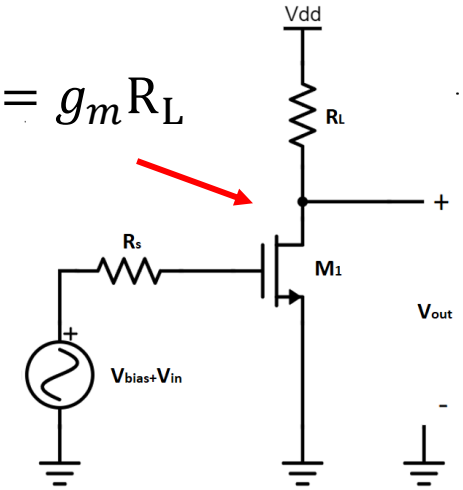
Solution 2b



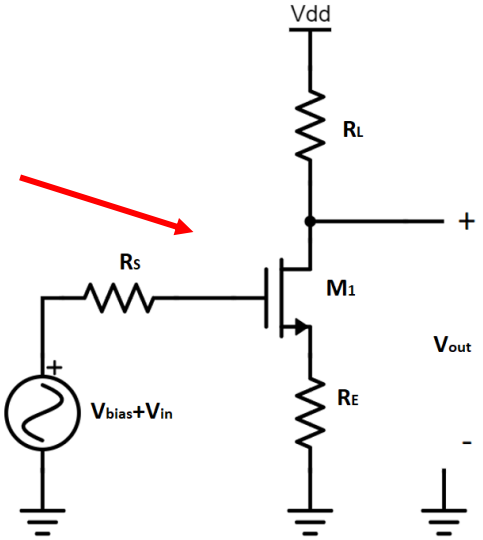
- $g_m V_{R_E} + \frac{V_{RE}}{R_E} = i_n$
- $V_{R_E} = \frac{i_n R_E}{1+g_m R_E}$
- $i_L = -\frac{i_n}{1+g_m R_E}$
- $\overline{dv_{n,M1,out}^2} = i_L^2 R_L^2 = \frac{4kT\gamma g_m R_L^2}{(1+g_m R_E)^2}$
- Or analyze with voltage source instead:
- $\overline{dv_{n,M1,out}^2} = \frac{4kT\gamma}{g_m} A_v^2 = \frac{4kT\gamma g_m R_L^2}{(1+g_m R_E)^2}$

Solution 2b

$$A_v = g_m R_L$$

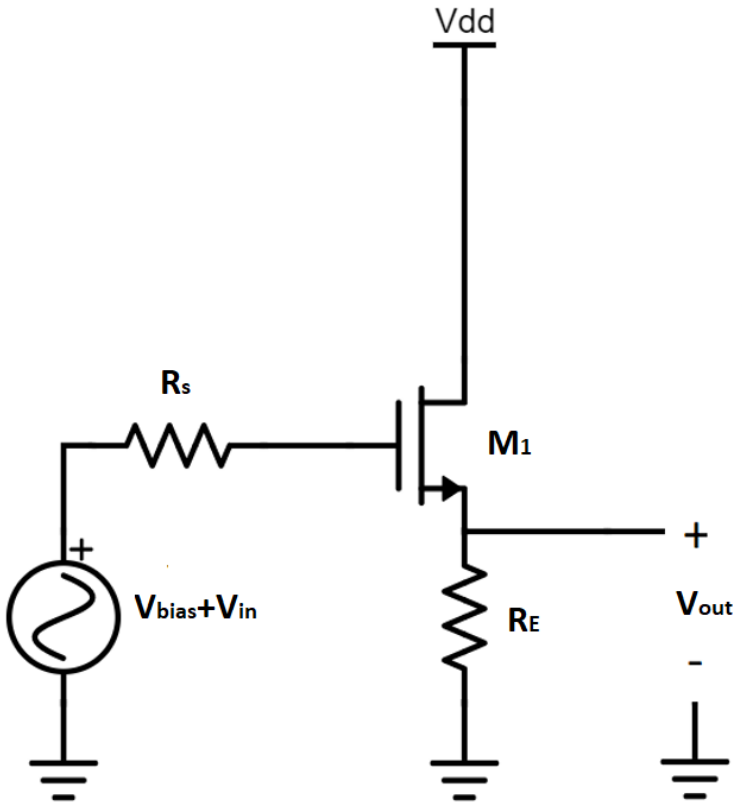


$$A_v = \frac{g_m R_L}{1 + g_m R_E}$$



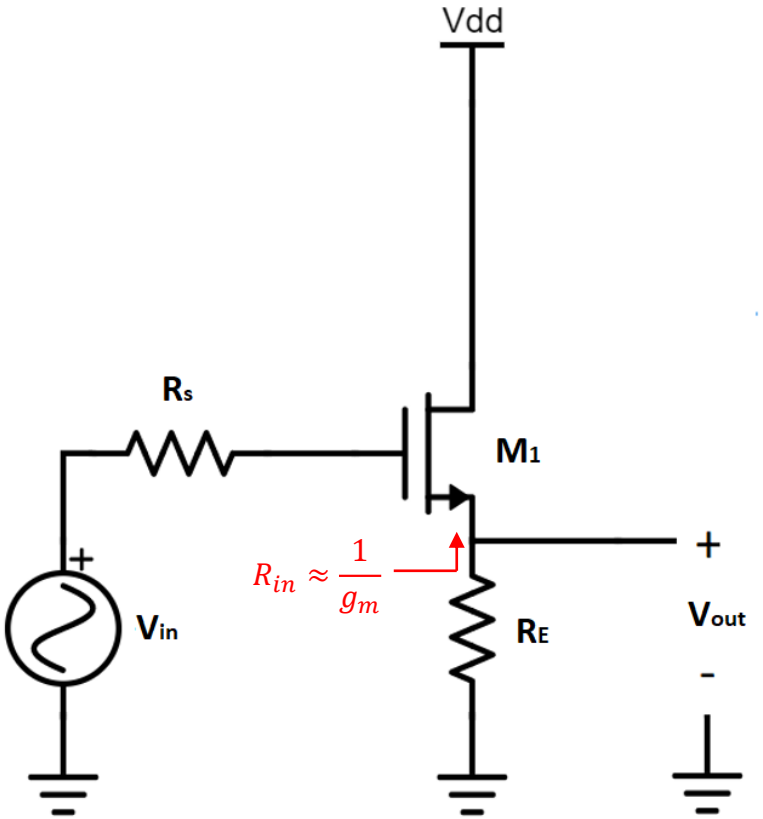
- $\overline{dv_{n,out}^2} = \overline{dv_{n,R_S,out}^2} + \overline{dv_{n,R_E,out}^2} + \overline{dv_{n,M1,out}^2} + \overline{dv_{n,R_L,out}^2}$
- $\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,out}^2}}{A_v^2} = \left[4kT(R_s + R_E) + \frac{4kT\gamma}{g_m} + \frac{4kTR_L}{A_v^2}\right] df$
- R_E contributes extra noise compared to common source.
- R_L noise contribution increases compared to common source.

Exercise 2c



- Calculate output and input referred noise of the common drain amplifier in the figure?
- Assume transistor with infinite output impedance and trans-conductance gm. Consider only thermal noise for calculations.

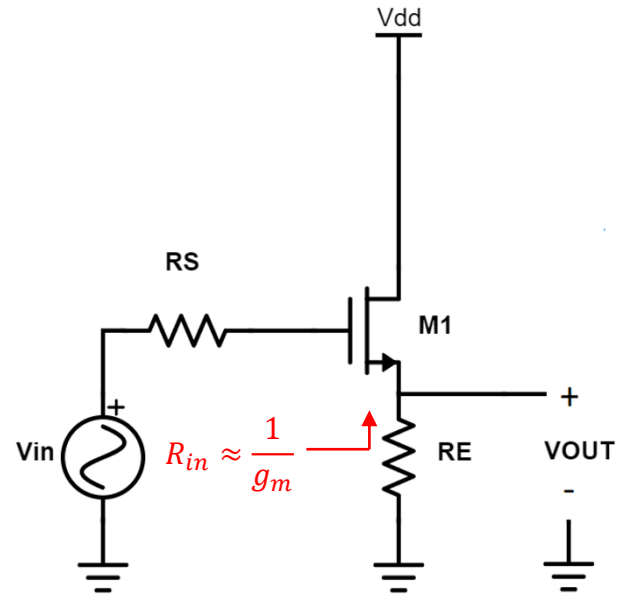
Solution 2c



Gain calculation

- $g_m(V_{in} - V_{out}) = \frac{V_{R_E}}{R_E}$
- $V_{out} = \frac{g_m R_E V_{in}}{1 + g_m R_E}$
- $A_v = \frac{g_m R_E}{1 + g_m R_E}$

Solution 2c



$$\overline{dv^2_{n,R_S,out}} = 4\mathrm{kTR_L}(\mathrm{A_v})^2 df$$

$$\overline{dv^2_{n,M1,out}} = \frac{4\mathrm{kT}\gamma}{g_m}(\mathrm{A_v})^2 df$$

$$\overline{dv^2_{n,R_E,out}} = 4\mathrm{kTR_E}\left(\frac{1/g_m}{1/g_m+R_E}\right)^2 df = 4\mathrm{kTR_E}\left(\frac{1}{1+g_m R_E}\right)^2 df$$

Resistor $\mathrm{R_E}$ noise attenuated by a factor of $\frac{1}{1+\mathrm{gmR_E}}$

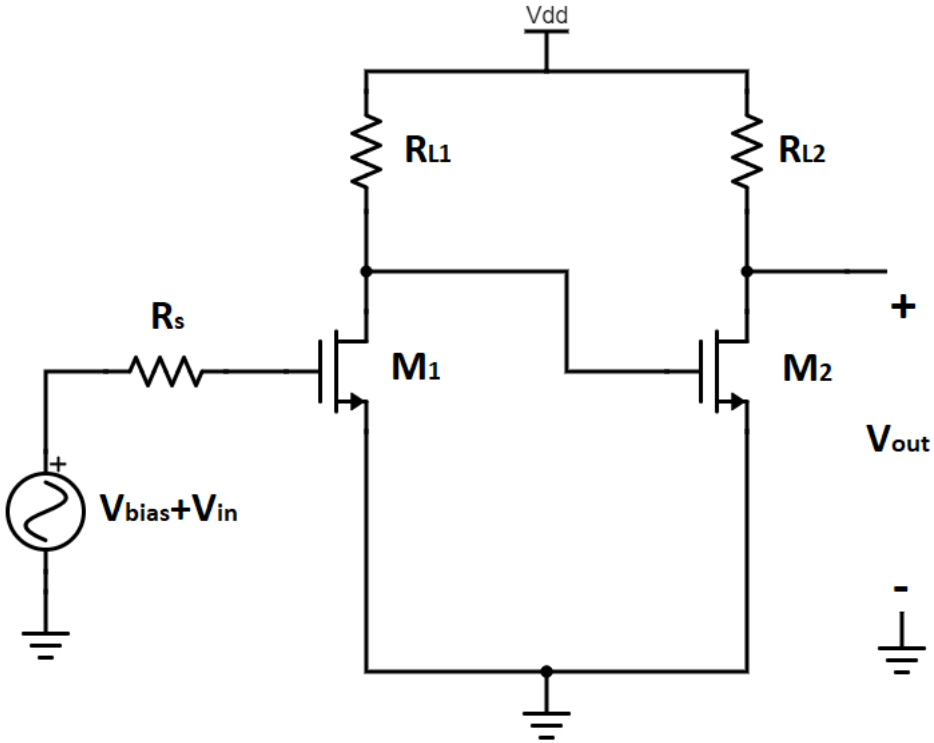
$$\overline{dv^2_{n,out}} = \overline{dv^2_{n,R_S,out}} + \overline{dv^2_{n,R_E,out}} + \overline{dv^2_{n,M1,out}}$$

$$\overline{dv^2_{n,in,eq}} = \frac{\overline{dv^2_{n,out}}}{A_v^2} = \left[4\mathrm{kTR_s} + \frac{4kTR_E}{(g_m R_E)^2} + \frac{4\mathrm{kT}\gamma}{g_m}\right] df$$

Session 4: Noise

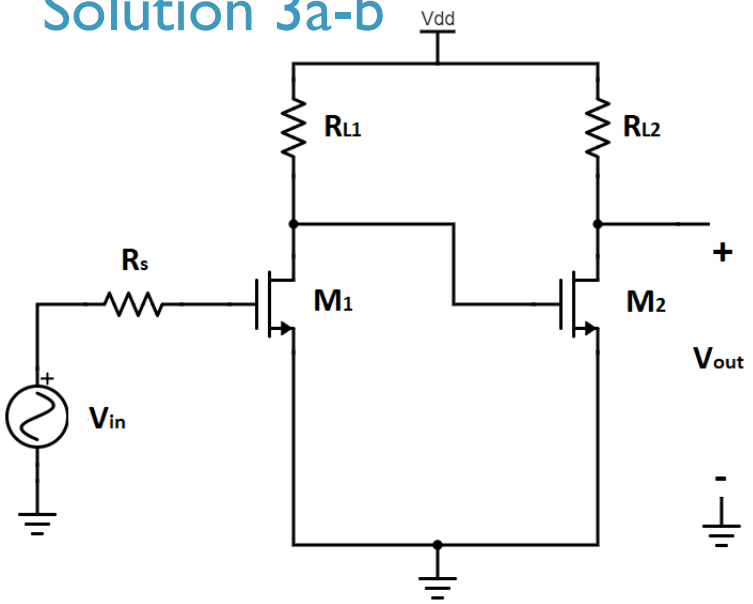
- Solve exercise 3
  - ETA: 10 min

Exercise 3a



- Calculate the input referred noise of the cascaded common source amplifier as shown in the figure. ? The transistors M1 and M2 have trans-conductance gm_1 and gm_2. Assume only thermal noise and neglect the flicker noise of transistors.
- Also assume infinite output impedance of the transistors with all transistors in saturation.

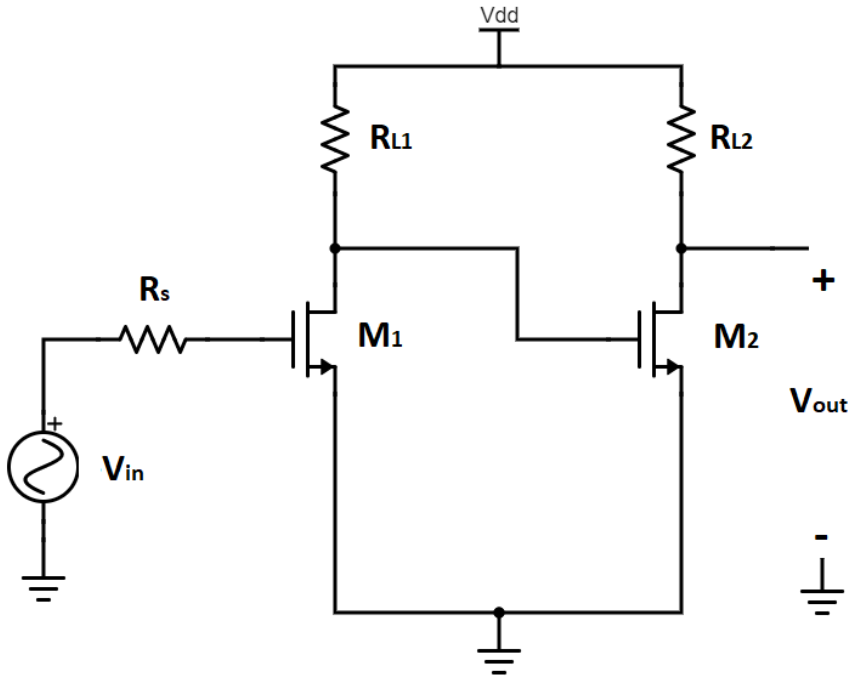
Solution 3a-b



- $A_{v1} = g_{m1} R_{L1}$ and $A_{v2} = g_{m2} R_{L2}$
- $\overline{dv_{n,R_s,out}^2} = 4kTR_s \, (A_{v1} A_{v2})^2 df$
- $\overline{dv_{n,M1,out}^2} = \frac{4kT\gamma}{g_{m1}} (A_{v1} A_{v2})^2 df$
- $\overline{dv_{n,R_{L1},out}^2} = 4kTR_{L1} \, (A_{v2})^2 df$
- $\overline{dv_{n,M2,out}^2} = \frac{4kT\gamma}{g_{m2}} (A_{v2})^2 df$
- $\overline{dv_{n,R_{L2},out}^2} = 4kTR_{L_2} df$

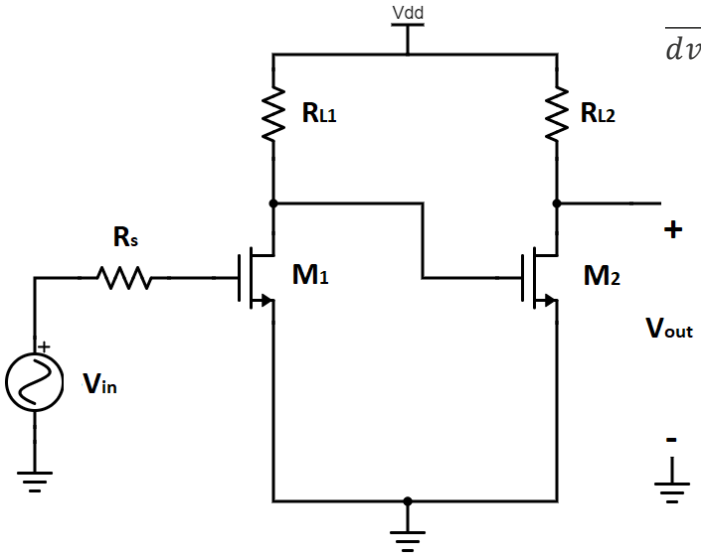
- $\overline{dv_{n,out}^2} = \overline{dv_{n,R_s,out}^2} + \overline{dv_{n,M1,out}^2} + \overline{dv_{n,R_{L1},out}^2} + \overline{dv_{n,M2,out}^2} + \overline{dv_{n,R_{L2},out}^2}$
- $\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,out}^2}}{A_{v1}^2 A_{v2}^2} = \left[4kTR_s + \frac{4kT\gamma}{g_{m1}} + \frac{4kTR_{L1}}{(A_{v1})^2} + \frac{4kT\gamma}{g_{m2}(A_{v1})^2} + \frac{4kTR_{L2}}{(A_{v1}A_{v2})^2} \right] df$

Exercise 3b



- What are the noise sources contributing most of the noise?
- From the expression you will see that in order to decrease the noise contribution of the second stage you can increase the gain of the first stage. Is their any limit (Consider channel length modulation and hence finite output impedance)? Any technique you can try to overcome the limit ?

Solution 3b



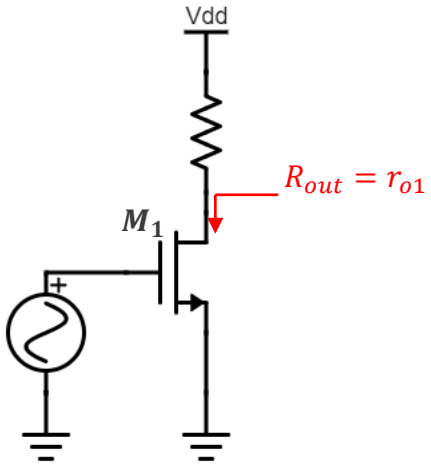
$$\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,out}^2}}{A_{v1}^2 A_{v2}^2} = \left[4\mathrm{kTR_s} + \frac{4\mathrm{kTg_{m1}}\gamma R_{L2}^2}{(\mathrm{A_{v1}})^2} + \frac{4\mathrm{kTR_{L1}}}{(\mathrm{A_{v1}})^2} + \frac{4\mathrm{kTg_{m2}}\gamma \mathrm{R_{L2}^2}}{(\mathrm{A_{v1}}A_{v2})^2} + \frac{4\mathrm{kTR_{L2}}}{(\mathrm{A_{v1}}\mathrm{A_{v2}})^2}\right] df$$

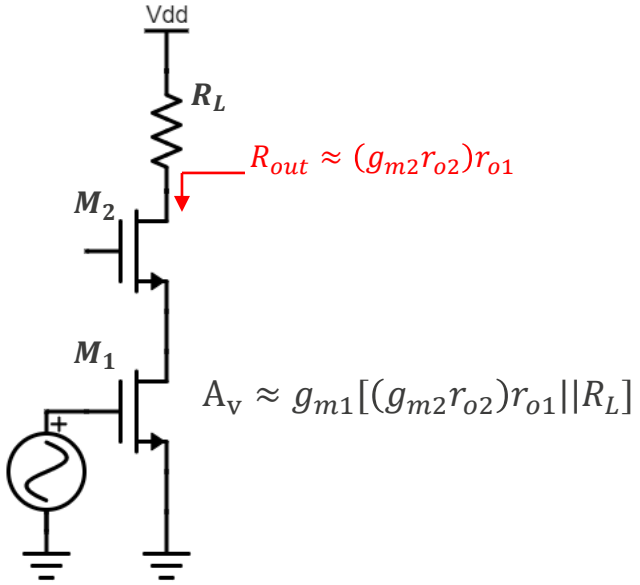
- Noise of second stage transistor suppressed by gain of the first stage.
- Hence, first stage noise usually dominates noise performance in cascaded amplifier.
- Increasing A_{v1} lower the noise contribution from the second stage
- What is the limit of A_{v1}?

Solution 3b

$$A_v = \frac{g_m R r_o}{R + r_o}$$

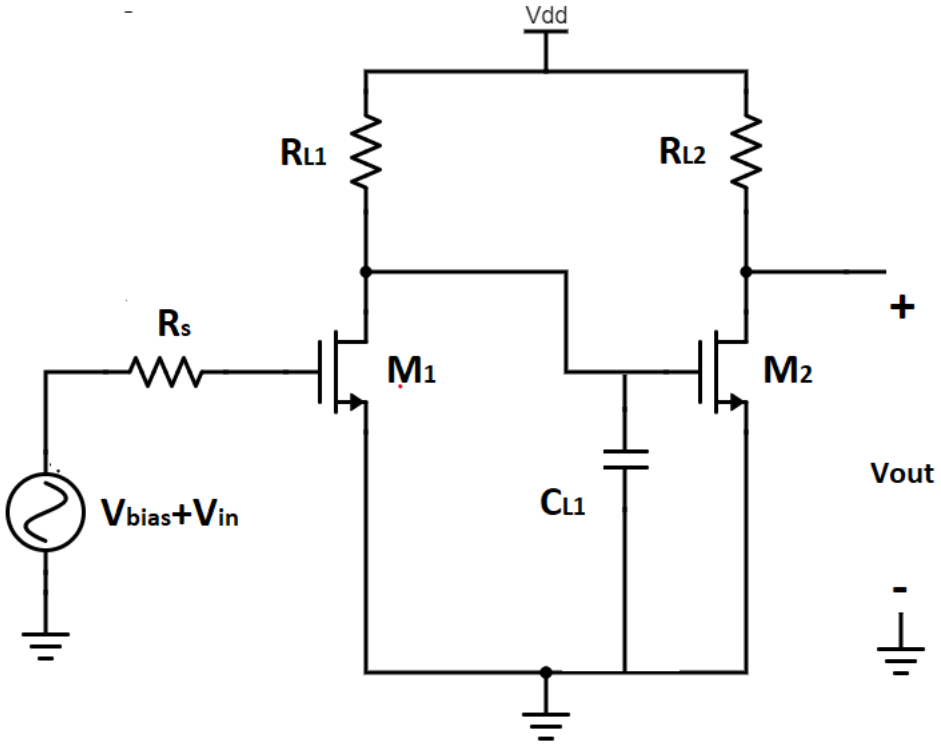
$$A_{vmax} = g_m r_o$$





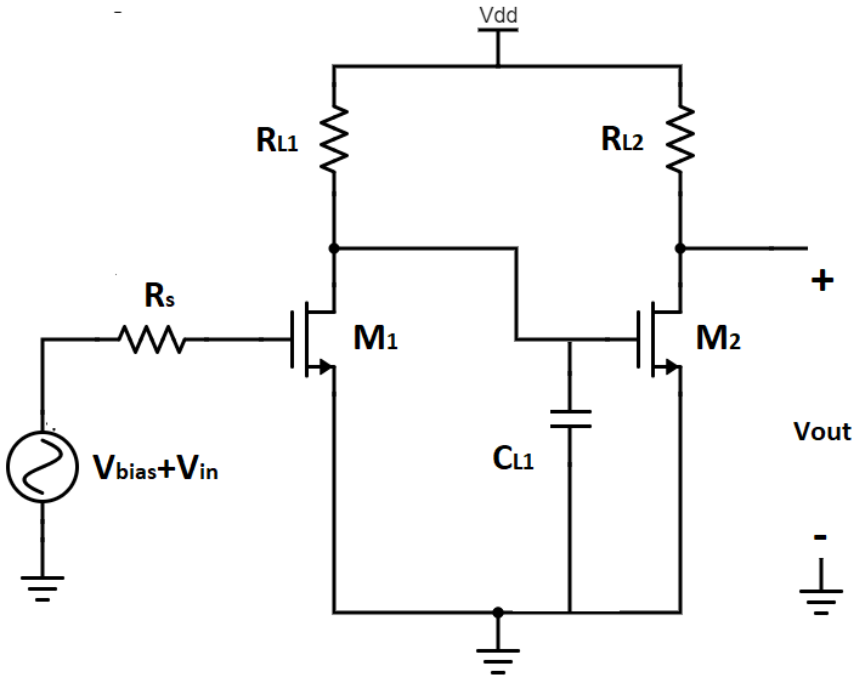
- First stage gain limited by finite output impedance or channel length modulation.
- Cascoding, can be used to increase first stage gain.

Exercise 3c



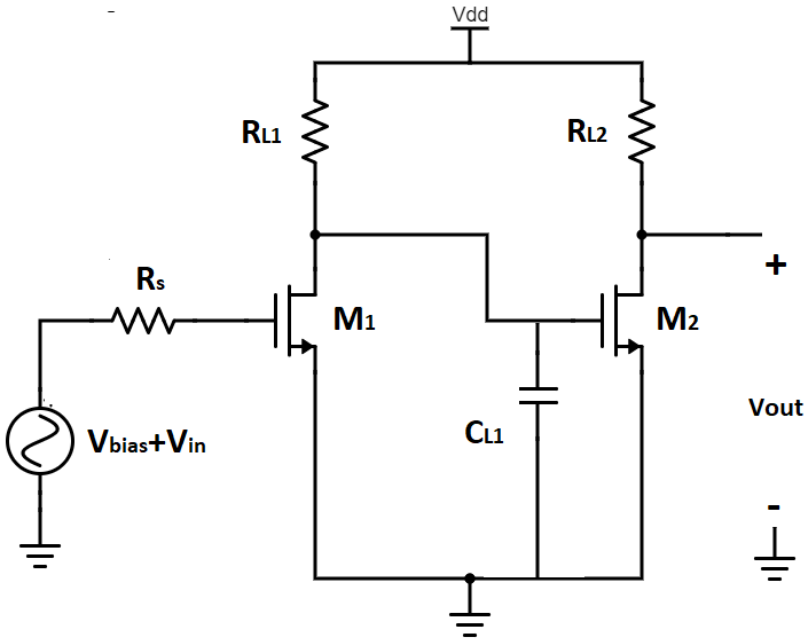
- Now add capacitance CL_1 to the output of the first stage. How will it change the input referred noise contribution of the second stage with frequency?

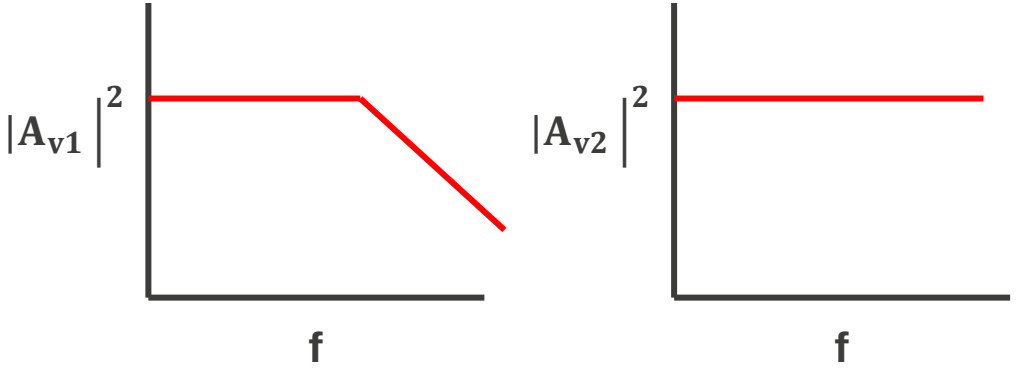
Solution 3c

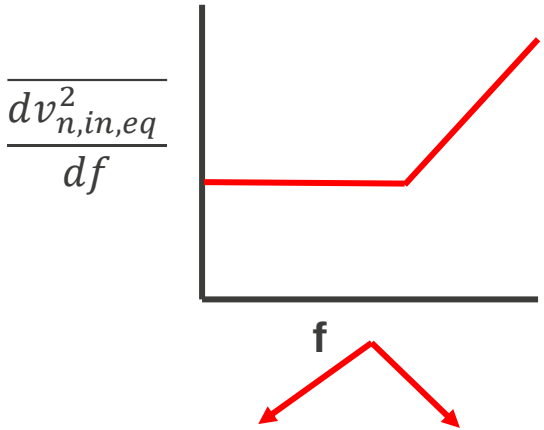


- $A_{v1} = -\frac{g_{m1}R_{L1}}{1+sC_{L1}R_{L1}}$, $|A_{v1}|^2 = \frac{g_{m1}^2 R_{L1}^2}{1+\omega^2 C_{L1}^2 R_{L1}^2}$
- $A_{v2} = -g_{m2}R_{L2}$
- $\overline{dv_{n,R_s,out}^2} = 4kTR_s|A_{v1}|^2|A_{v2}|^2$
- $\overline{dv_{n,M1,out}^2} = \frac{4kT\gamma}{g_{m1}}|A_{v1}|^2|A_{v2}|^2$
- $\overline{dv_{n,R_{L1},out}^2} = \frac{4kTR_{L1}}{1+\omega^2 C_{L1}^2 R_{L1}^2}|A_{v2}|^2$
- $\overline{dv_{n,M2,out}^2} = \frac{4kT\gamma}{g_{m1}}|A_{v2}|^2$
- $\overline{dv_{n,R_{L2},out}^2} = 4kTR_{L2}$
- $\overline{dv_{n,out}^2} = \overline{dv_{n,R_s,out}^2} + \overline{dv_{n,M1,out}^2} + \overline{dv_{n,R_{L1},out}^2} + \overline{dv_{n,M2,out}^2} + \overline{dv_{n,R_{L2},out}^2}$

Solution 3c







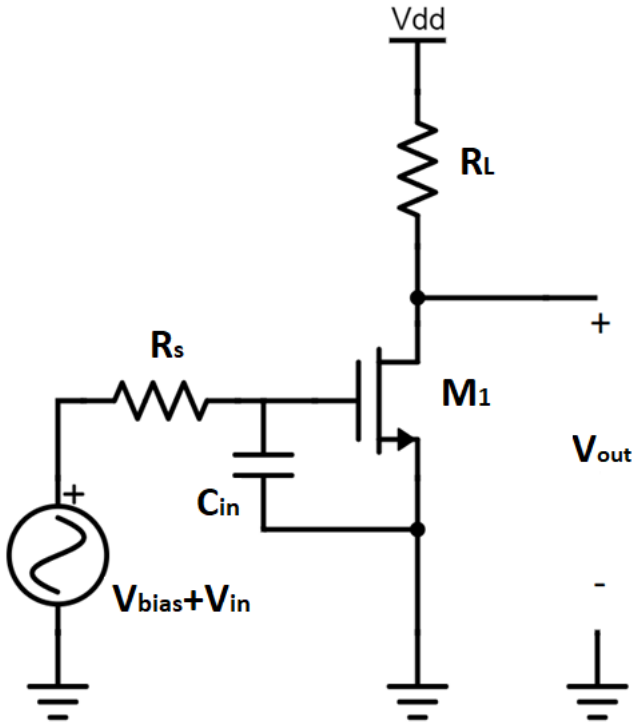
- $$\overline{dv^2_{n,in,eq}} = \frac{\overline{dv^2_{n,out}}}{(|A_{v1}||A_{v2}|)^2} = 4kTR_s + \frac{4kT\gamma}{g_{m1}} + \frac{4kTR_{L1}}{(g_{m1}R_{L1})^2} + \frac{4kT\gamma}{g_{m2}(|A_{v1}|)^2} + \frac{4kTR_{L2}}{(|A_{v1}||A_{v2}|)^2}$$

Refer to lecture note Fig. 4.7.2

Session 4: Noise

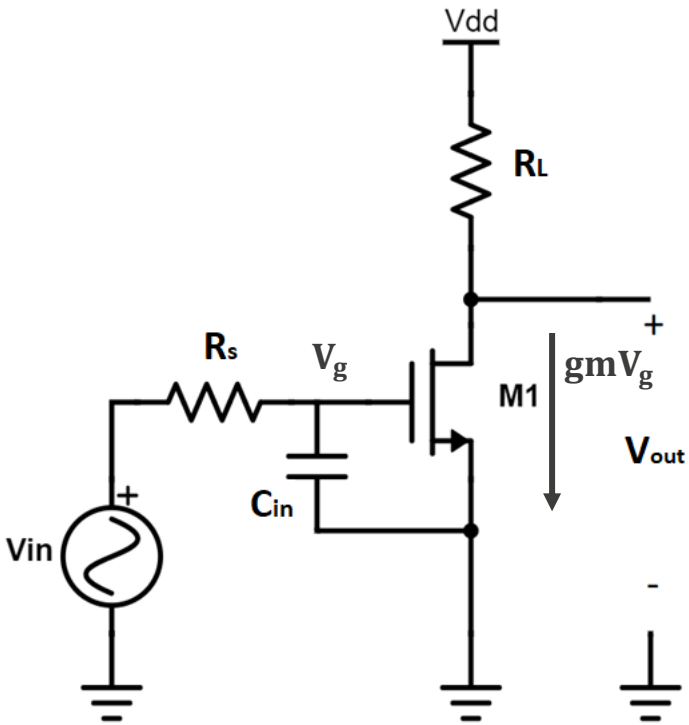
- Solve exercise 4
  - ETA: 10 min

Exercise 4a



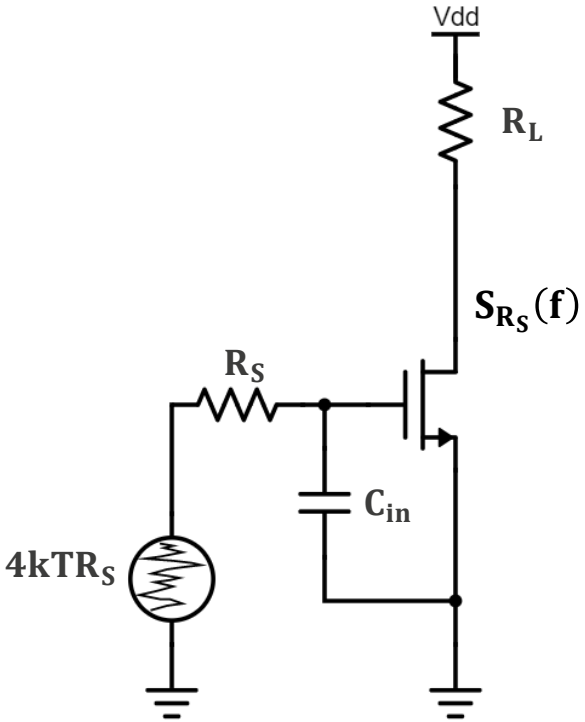
- Figure shows common source amplifier with total input capacitance C_{in}. In the figure, R_L=5kΩ, R_S=100kΩ , C_{in}=33fF and gm=1mS.
- For transistor M1 drain flicker noise current spectral density at 1Hz=$5 * 10^{-15}$ A^2/Hz.
- Plot the output and input referred noise voltage spectral density with frequency in MATLAB. In this question consider both thermal and flicker noise contribution ? Take γ=2/3.

Solution 4a

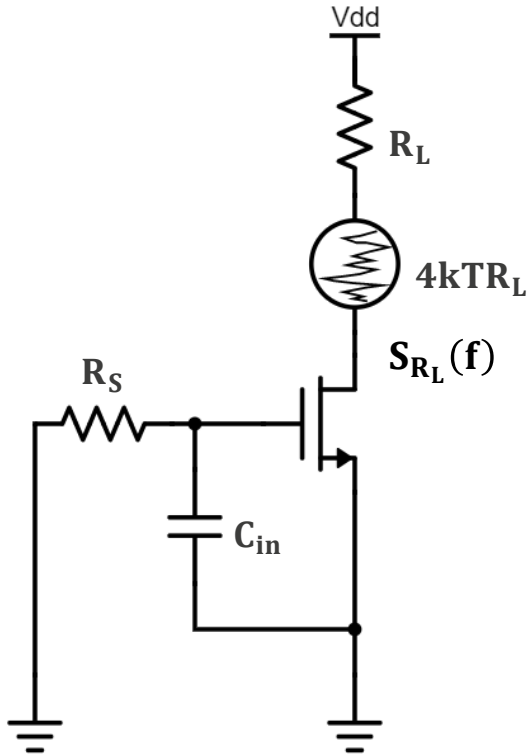


- $V_g = \frac{V_{in}}{1+sC_{in}R_s}$
- $V_{out} = -g_m V_g R_L$
- $V_{out} = \frac{-g_m R_L V_{in}}{1+sC_{in}R_s}$ thus $A_v = \frac{-g_m R_L}{1+sC_{in}R_s}$
- $|A_v|^2 = \frac{g_m^2 R_L^2}{1+\omega^2 C_{in}^2 R_S^2}$

Solution 4a

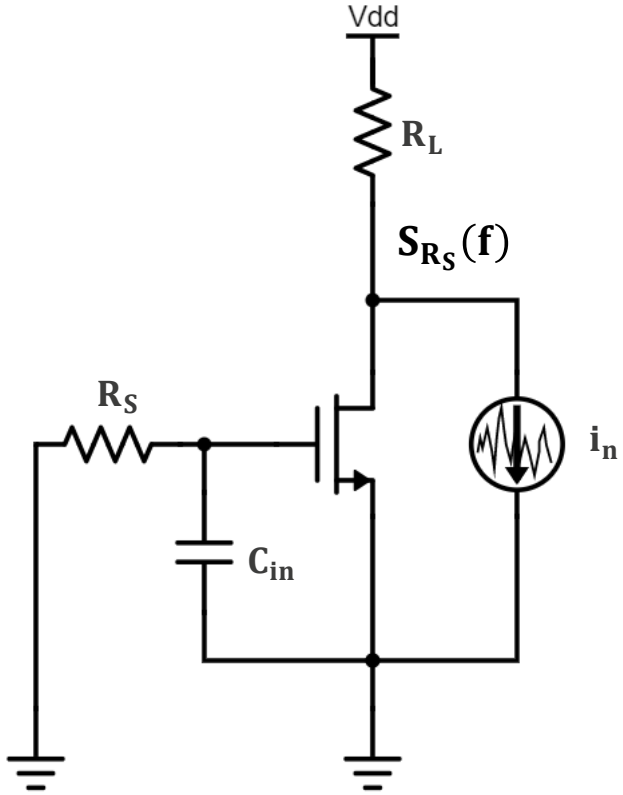


- $\overline{dv_{n,R_s,out}^2} = 4\mathrm{kTR_s}\ (|\mathrm{A_v}|)^2 df$



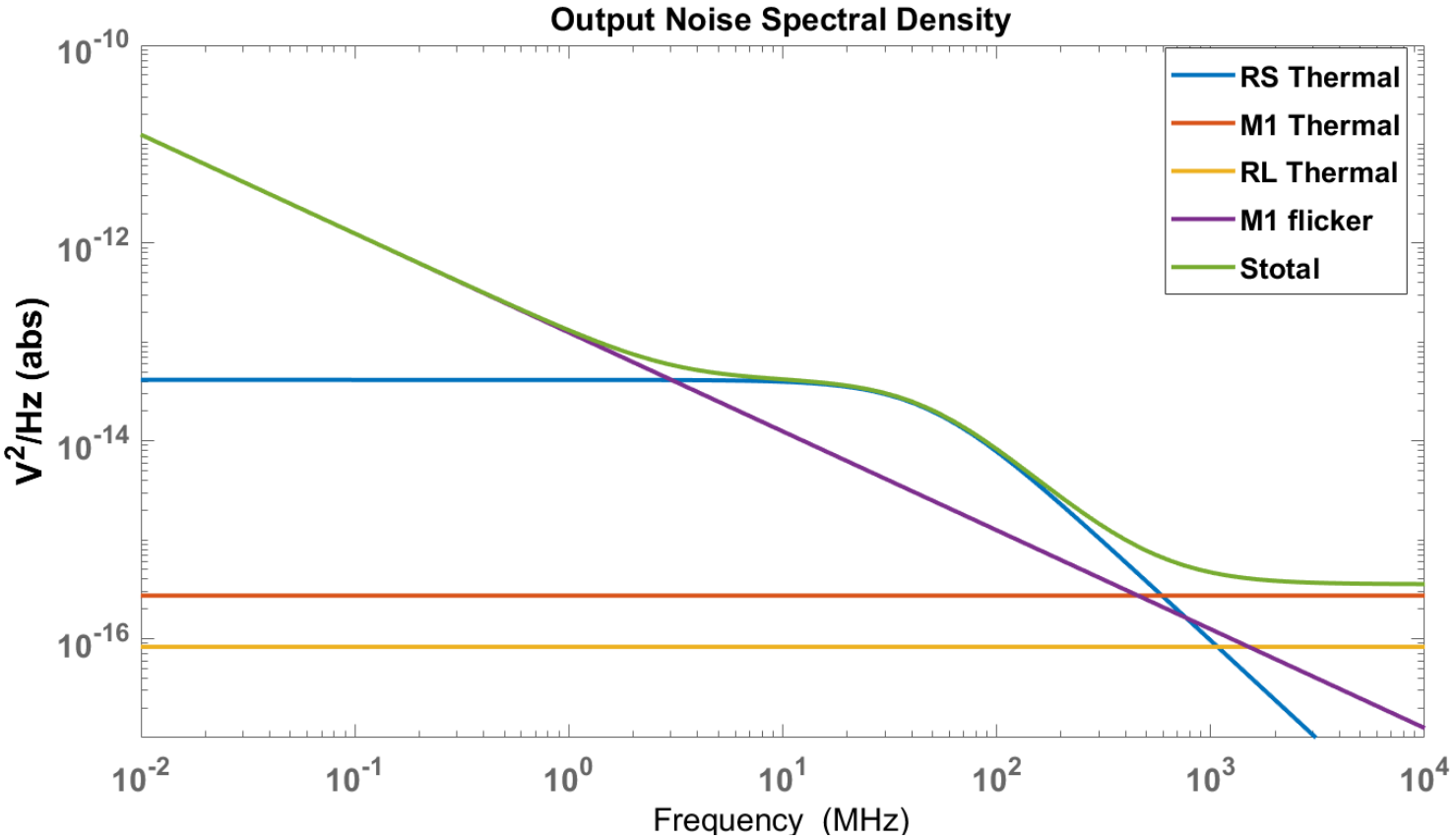
- $\overline{dv_{n,R_L,out}^2} = 4\mathrm{kTR_L} df$

Solution 4a

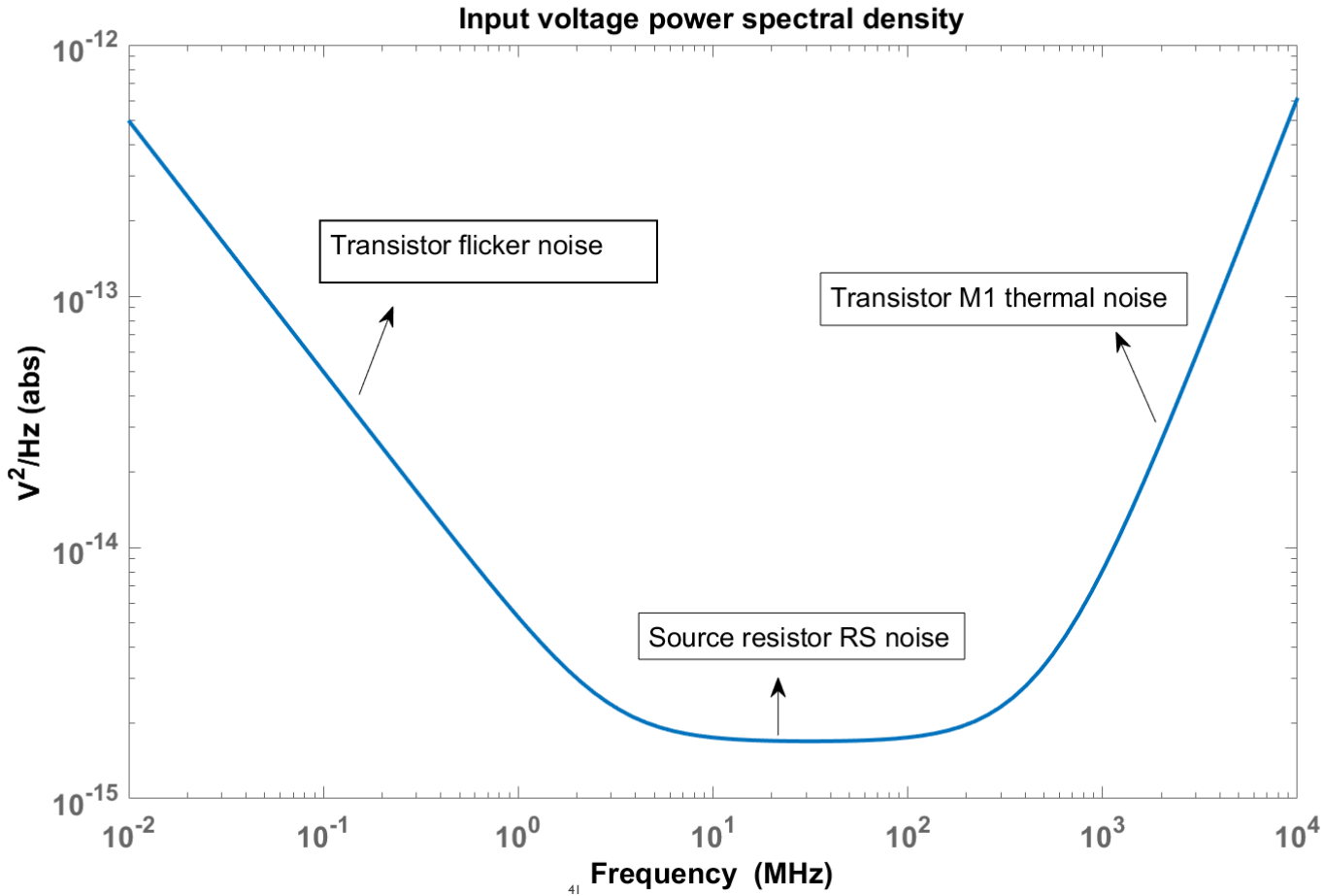


- $\overline{dv_{n,R_S,out}^2} = i_n^2 R_L^2 = 4kT\gamma g_m R_L^2$
- $\overline{dv_{n,out,thermal}^2} = [4kTR_s(|A_v|)^2 + 4kT\gamma g_m R_L^2 + 4kTR_L]df$
- $\overline{dv_{n,out,flicker}^2} = i_f^2 R_L^2 df$ where $i_f^2 = \frac{5*10^{-15}}{f}\frac{A^2}{Hz}$
- $\overline{dv_{n,out}^2} = \overline{dv_{n,out,thermal}^2} + \overline{dv_{n,out,flicker}^2}$
- $\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,out}^2}}{|A_v^2|}$
- Using the values given in the question we can plot the noise contribution.

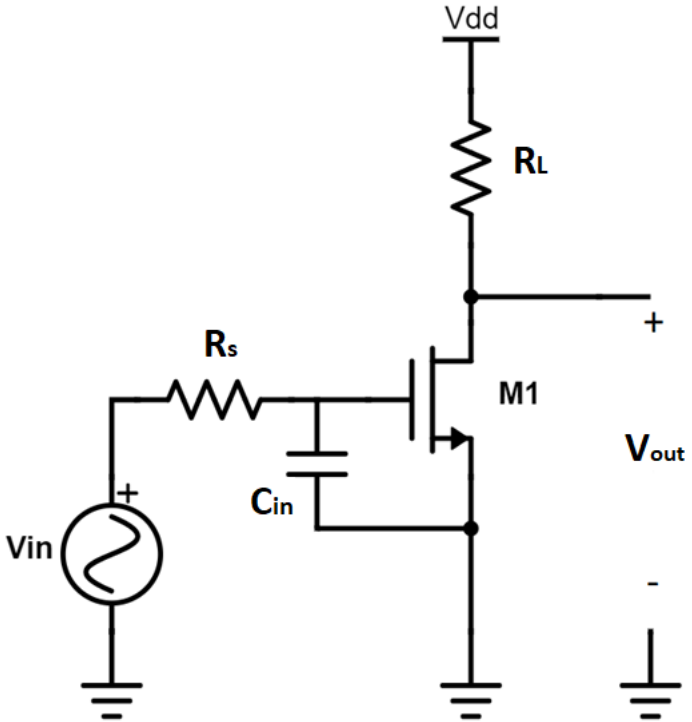
Solution 4a


Output Noise Spectral Density
RS Thermal
M1 Thermal
RL Thermal
M1 flicker
Stotal
V2/Hz (abs)
10-10
10-12
10-14
10-16
10-2
10-1
100
101
102
103
104
Frequency (MHz)

Solution 4a


Input voltage power spectral density
Transistor flicker noise
Transistor M1 thermal noise
Source resistor RS noise
V²/Hz (abs)
Frequency (MHz)
10^-12
10^-13
10^-14
10^-15
10^-2
10^-1
10^0
10^1
10^2
10^3
10^4

Exercise 4b



- The input signal bandwidth is from 10MHz to 100MHz ? Calculate the input referred integrated noise in the given bandwidth and minimum detectable signal value ?

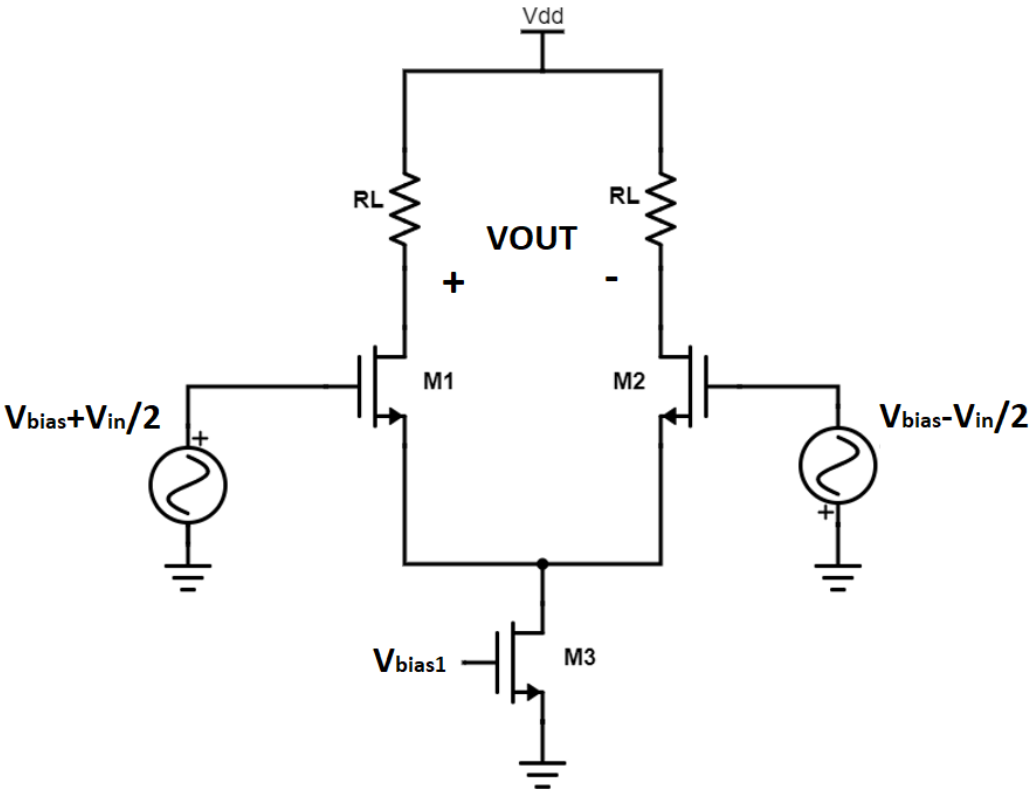
Solution 4b

- $\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,out}^2}}{|A_v^2|}$ and $|A_v|^2 = \frac{g_m^2 R_L^2}{1+\omega^2 C_{gs}^2 R_S^2}$
- Input referred noise dominated by source resistor R_S.
- Source resistor noise $4kTR_S = 1.65 * 10^{-15} \frac{V^2}{Hz}$
- Input referred integrated noise in 90MHz band $\overline{V_n^2}=4kTR_SB=1.5 * 10^{-7}\ V^2$
- Rms integrated noise $= \sqrt{1.5 * 10^{-7}}$=0.39mV rms which is minimum detectable signal.

Session 4: Noise

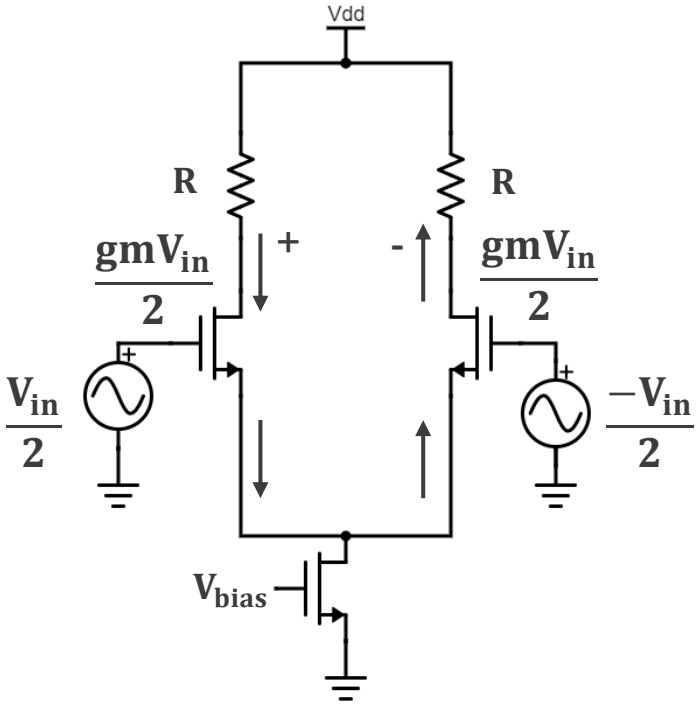
- Solve exercise 5
  - ETA: 15 min

Exercise 5a

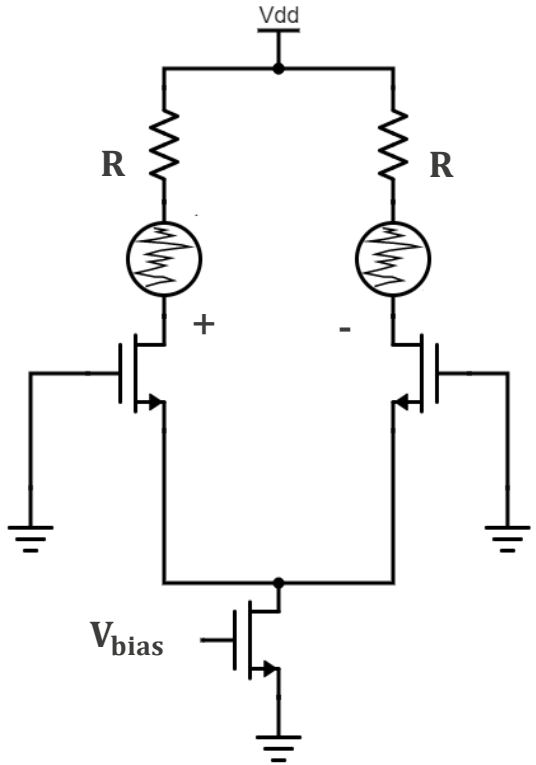


- Figure shows a different pair. Transistor M1 and M2 have trans-conductance gm. Assuming bias current from transistor M3 does not add any noise. What is total input referred noise of the differential pair ? Consider only thermal noise.

Solution 5a

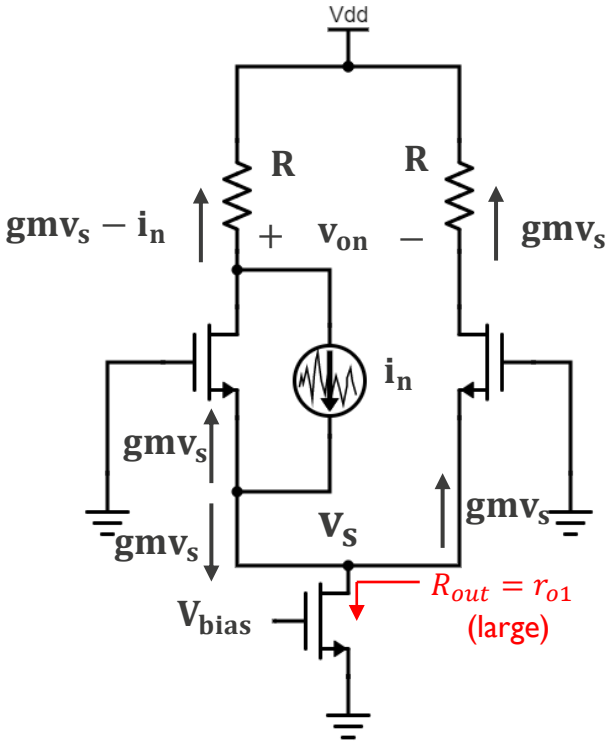


- $A_v = -g_m R$



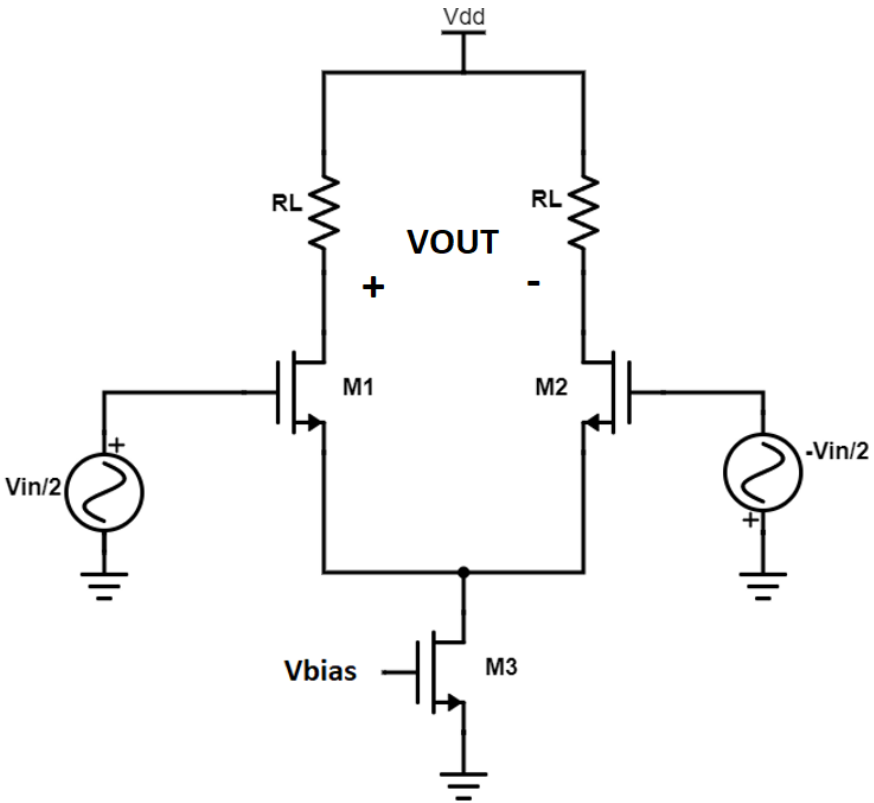
- $\overline{dv_{n,R,out}^2} = [4kTR + 4kTR]df$

Solution 5a



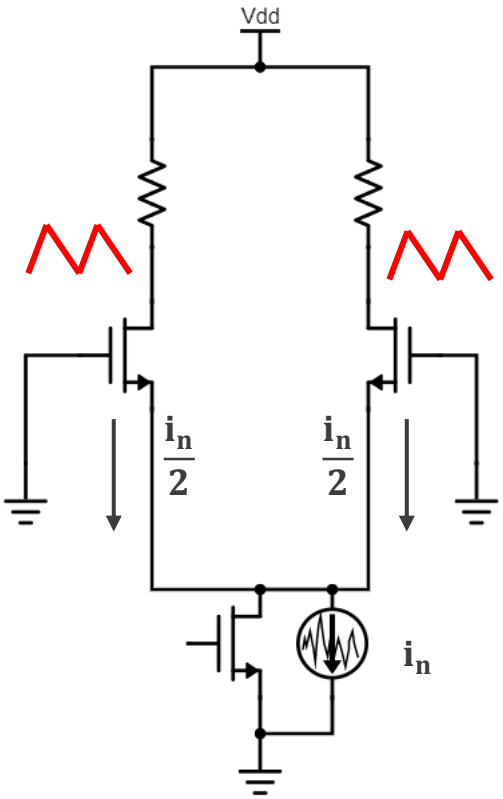
- $2g_m v_s = i_n$ thus $gmv_s = \frac{i_n}{2}$
- $v_{on} = (g_m v_s - i_n)R \; - g_m v_s R = -i_n R$
- Hence, $\overline{dv^2_{n,M1,out}} = \overline{dv^2_{n,M2,out}} = i_n^2 R^2 = 4kT\gamma gmR^2$
- $\overline{dv^2_{n,out}} = [8kT\gamma g_m R^2 + 8kTR]df$
- $\overline{dv^2_{n,in,eq}} = \frac{\overline{dv^2_{n,out}}}{g_m^2 R^2} = \left[\frac{8kT\gamma}{g_m} + \frac{8kTR}{g_m^2 R^2}\right] df$

Exercise 5b-c



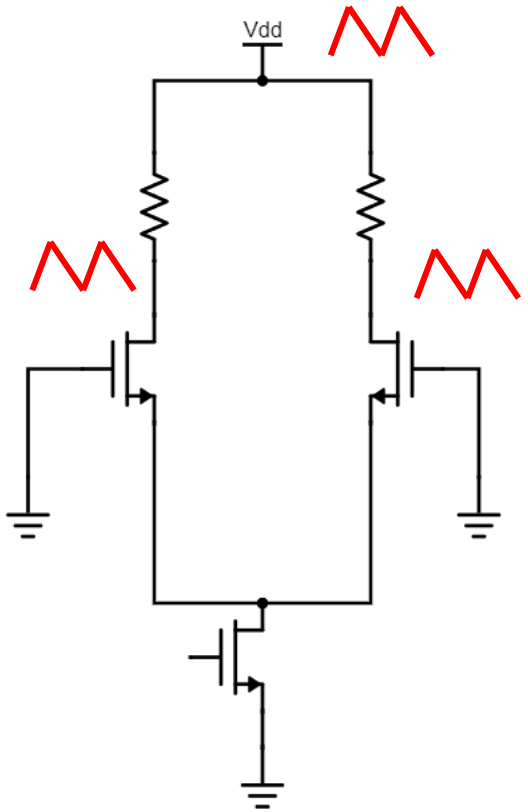
- Now imagine transistor M3 adds noise. Do you think it will add noise to the differential output? Can flicker noise in M3 change the gain of the amplifier with time ?
- What do you think will be the impact of noise in supply Vdd ?

Solution 5b



- Noise added by M_3 common mode are cancelled at the output if the input signal is small. Some exceptions exists like Mixers and Oscillators.
- Flicker noise in M_3 can be thought of as very slow fluctuating current.
- Hence flicker noise in M_3 will slowly vary g_m of differential pair and hence the gain.

Solution 5c

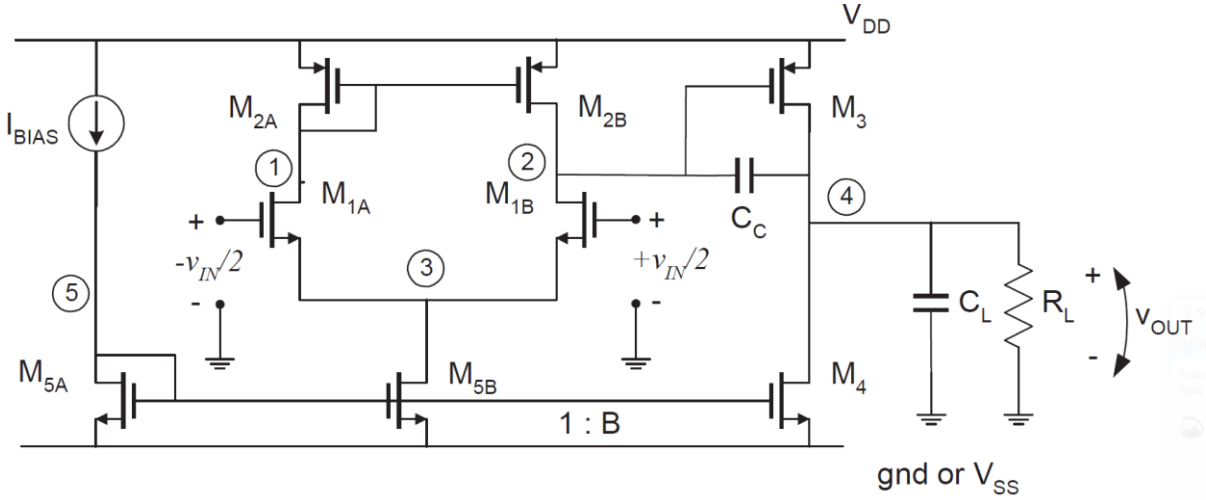


- Fluctuations in V_{dd} are common mode, also cancelled at the output.

Session 4: Noise

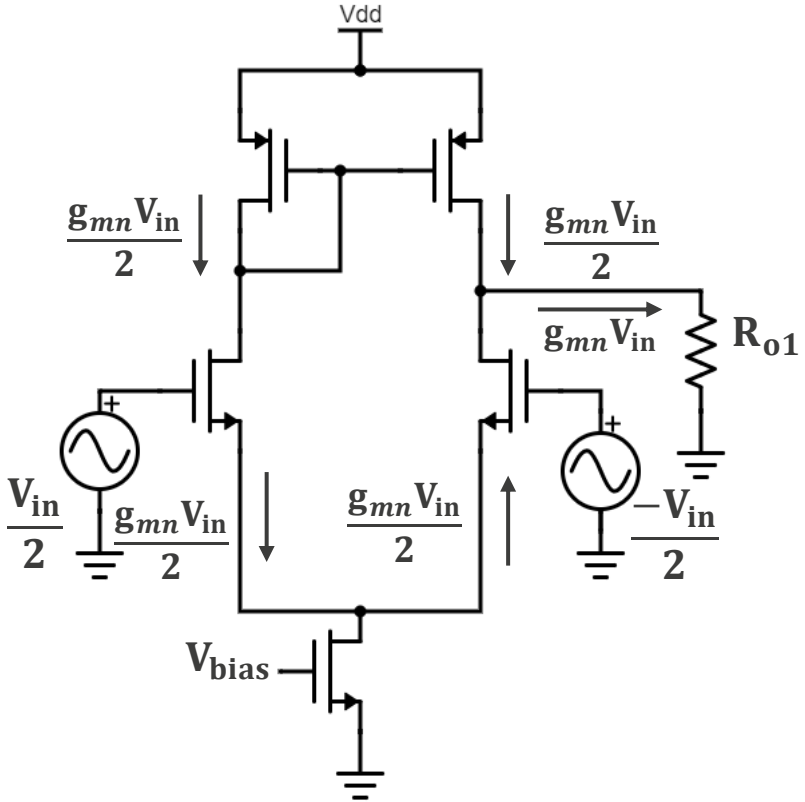
- Solve exercise 6
  - ETA: 10 min

Exercise 6a



- Calculate the input referred noise contribution at low frequencies from the transistors M_{1A} , M_{1B} , M_{2A} and M_{2B}? Assume the trans-conductance of transistors M_{1A} and M_{1B} is gm_n and trans-conductance of M_{2A} and M_{2B} is gm_p.
- Consider only thermal noise of the transistors. Low frequency means frequencies much less that the dominant pole.

Solution 6



- $A_{V1} = -g_{mn} R_{o1}$

A_{v1}: gain of the first stage

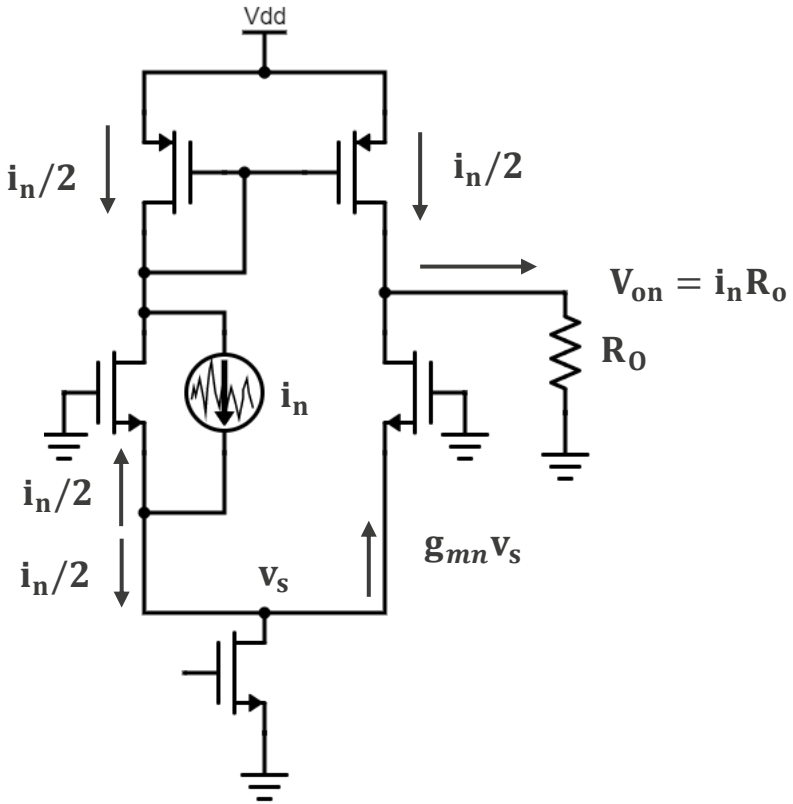
R_{o1}: output impedance of the first stage

No need to consider second stage to calculate the input-referred noise contribution from the first stage:

$$\overline{dv_{n,out}^2} = A_{v2}^2 \overline{dv_{n,1,out}^2}$$

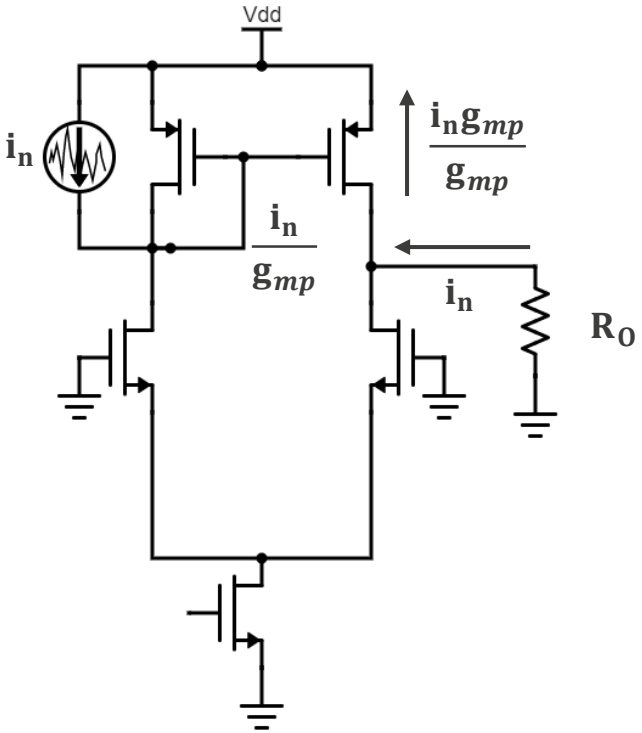
$$\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,out}^2}}{A_{v1}^2 A_{v2}^2} = \frac{\overline{dv_{n,1,out}^2}}{A_{v1}^2}$$

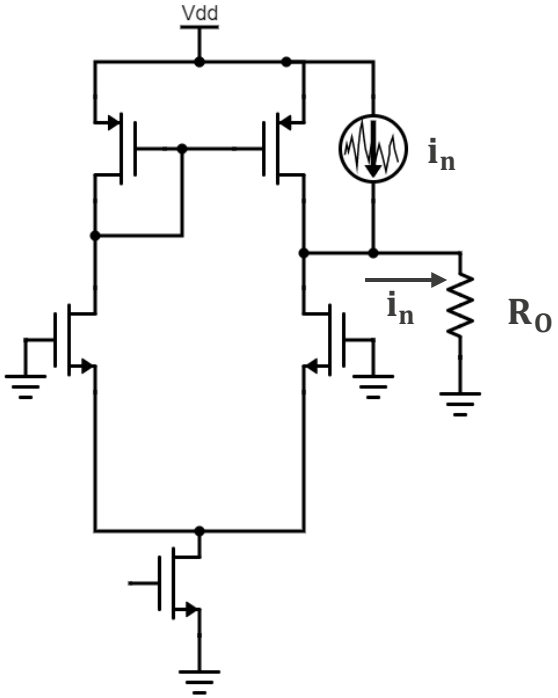
Solution 6a



- $V_{on} = i_n R_o$
- $i_n^2 = 4kT\gamma g_m$
- $\overline{dv_{n,M1A,out}^2} = i_n^2 R_o^2 df = 4kT\gamma g_{mn} R_o^2 df$
- Similarly $\overline{dv_{n,M1B,out}^2} = 4kT\gamma g_{mn} R_o^2 df$

Solution 6a



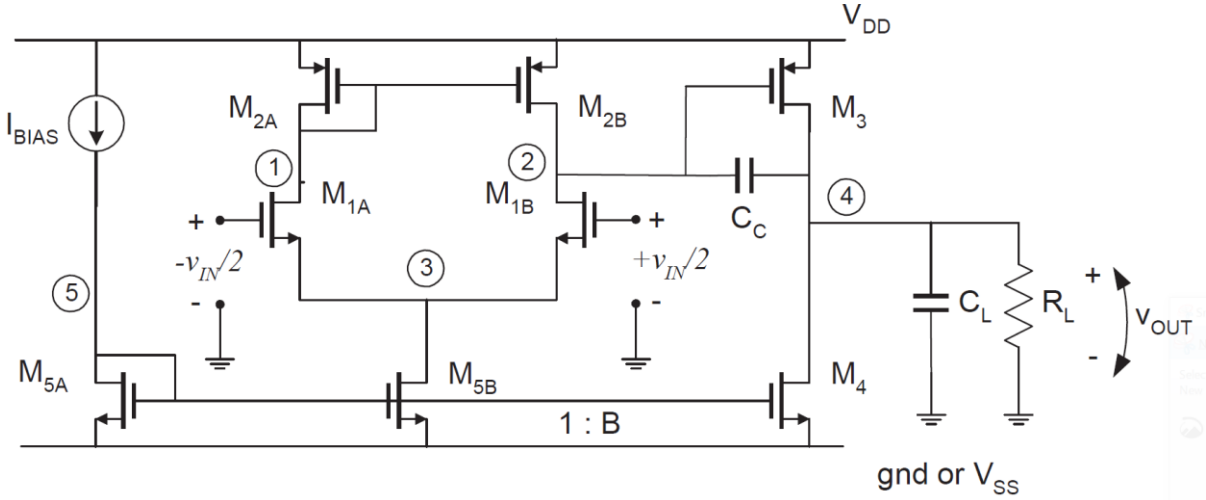


- $\overline{dv_{n,M2A,out}^2} = 4kT\gamma g_{mp} R_o^2 df$
- $\overline{dv_{n,M2B,out}^2} = 4kT\gamma g_{mp} R_o^2 df$

Solution 6a

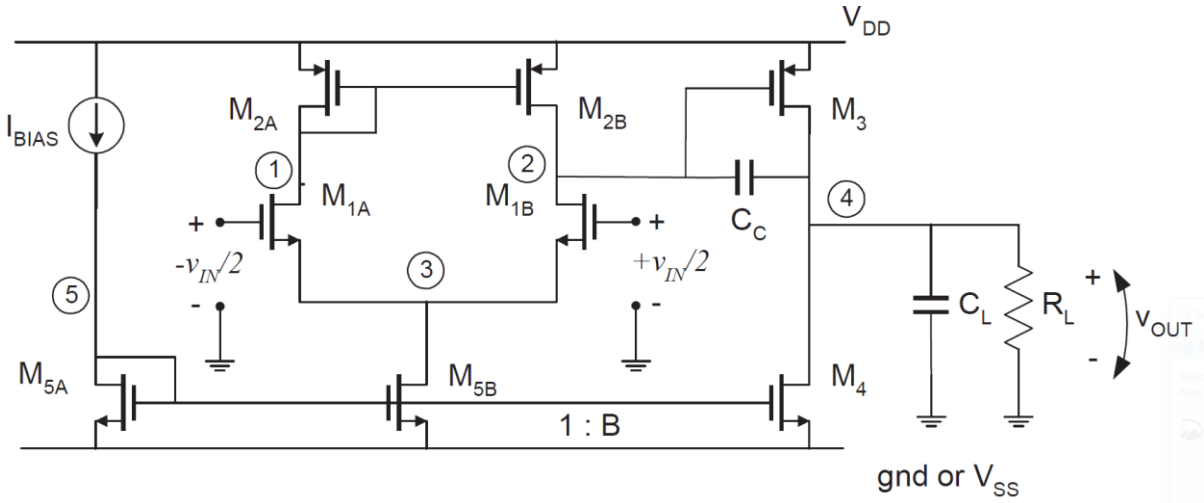
- $\overline{dv_{n,out}^2} = \overline{dv_{n,M1A,out}^2} + \overline{dv_{n,M1B,out}^2} + \overline{dv_{n,M2A,out}^2} + \overline{dv_{n,M2B,out}^2}$
- $\overline{dv_{n,in,eq}^2} = \frac{\overline{dv_{n,M1A,out}^2}}{A_{v1}^2} = [\frac{8kT\gamma}{g_{mn}} + \frac{8kT\gamma g_{mp}}{g_{mn}^2}]df$

Exercise 6b



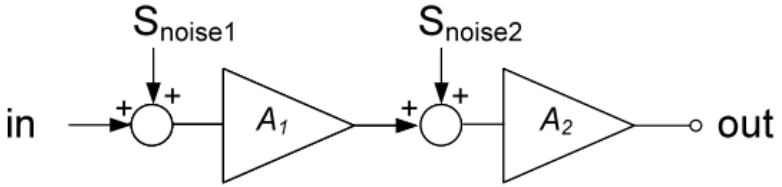
- Can you comment on noise contribution of transistors M_3 and M_4 at very low frequencies.
- At what frequency approximately will the 2nd stage noise will begin to dominate? Take low frequency output impedance of first and second as R_{o1} and R_{o2}. (Hint this frequency to first order is the frequency where the gain of the first stage approximately drops to 1)

Solution 6b



- Noise contribution of M_3 and M_4 suppressed by gain of first stage .
- Output gain of the first stage of the OTA = $\frac{g_{mn}R_{o1}}{1+sC_c g_{m3}R_{o1}R_{o2}}$
- $f(A_v=1)=\frac{g_{mn}}{2\pi g_{m3}C_c R_{o2}}$

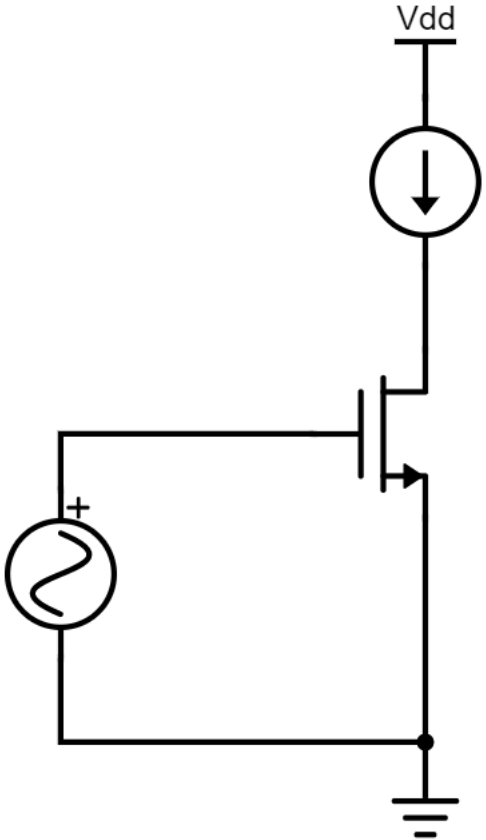
$$\overline{dv^2_{n,out,tot}} = \overline{dv^2_{n,in,stage1,eq}}\,A^2_{v1}A^2_{v2} + \overline{dv^2_{n,in,stage2,eq}}\,A^2_{v2}$$



MATLAB session

- Session 1: Fix V_{OV}, increment I_{BIAS} at each step, plot how the current/width/thermal noise/flicker noise changes with g_m
- Session 2: Finding the corner noise
- Session 3: Fix I_{BIAS}, increment V_{OV} at each step, plot how the width/thermal noise/flicker noise changes with g_m

MATLAB Session 1



Fix Vov, increment I_{BIAS} at each step, plot how the current/width/thermal noise/flicker noise changes with g_m

MATLAB Session I

```
KeepConstant = 'ids'; % {gm , ids}
Mn1.vov          = 0.15;
Mn1.lg           = 5*tableLmin(NRVT);
Mn1.vds          = 0.6;

   if strcmp(KeepConstant, 'gm')
       Mn1.gm       = 2e-3;
   else
       Mn1.ids      = 200e-6;
   end
```

- $I_{ds} = \frac{W}{2L}\mu_n C_{ox} V_{ov}^2$
- V_{ov}, V_{ds} and channel length is fixed.
- I_{ds} is varied by changing W.
- Vary I_{ds} from 50u to 1mA.
- Parameter KeepConstant ='ids' enables computation of channel width and hence gm based of Mn1.ids parameter as shown.

MATLAB Session 1

Accessing transistor noise from MATLAB

Thermal noise: $\overline{di^2_{d,thermal}}$ ⇒ Mn1.di2_id
Note: to calculate the integrated noise from f_1 to f_2, need to use Mn1.di2_id*($f_2 - f_1$)

Flicker noise: $\overline{di^2_{d,flicker}}$ ⇒ Mn1.di2_fn

Note: the flicker noise spectral density is Mn1.di2_fn divided by the frequency. Therefore, to calculate the integrated noise from f_1 to f_2, need to use Mn1.di2_fn*(log(f_2) – log(f_1))

MATLAB Session I

```
% Input referred noise spectral densities at frequency f1
d_vn2_thermal_in__Mn1_f1= d_vn2_thermal_in__Mn1;
d_vn2_flicker_in__Mn1_f1= d_vn2_flicker_in__Mn1/fnoise_spot1;
d_vn2_total_in__Mn1     = d_vn2_thermal_in__Mn1_f1+d_vn2_flicker_in__Mn1_f1;
```

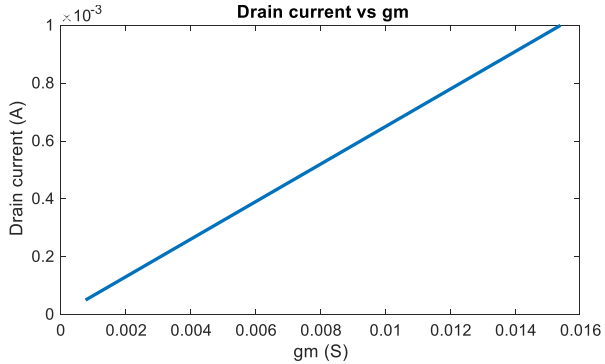
- Vary the drain current using Mn1.ids from 50u to 1mA .
- Plot drain current vs gm.
- Plot Transistor width vs gm.
- Also plot input referred thermal noise , flicker noise vs gm.
- Keep fnoise_spot1 parameter as 1Hz for now. It is the frequency at which spot noise is obtained.

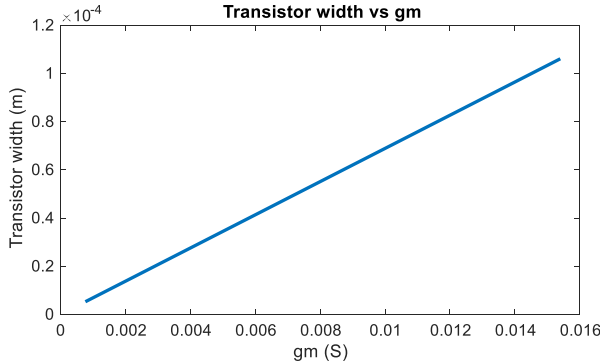
MATLAB Session I

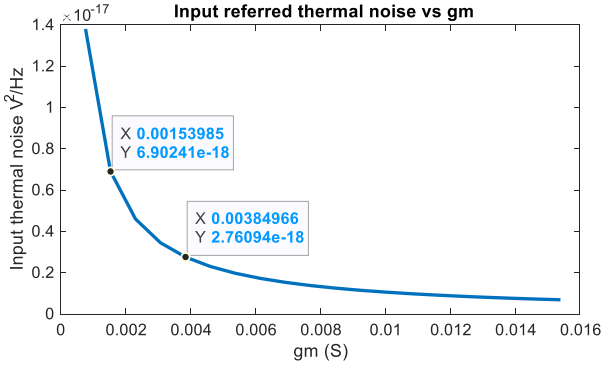
- $I_{ds} = \frac{W}{2L}\mu_n C_{ox} V_{ov}^2$ and hence $I_{ds} \alpha\ W$
- $gm = 2\frac{I_{ds}}{V_{ov}}$ or $gm = \frac{W}{L}\mu_n C_{ox} V_{ov}$
- $S_{thermal}(f) = \frac{4kT\gamma}{gm}$ and hence $S_{thermal}(f)\ \alpha \frac{1}{I_{ds}}$
- $S_{flicker}(f) = \frac{K}{C_{ox} W L f}$ and here f=1Hz .
- To decrease input referred thermal noise by a factor of 2 keeping same V_{ov} , both I_{ds} and hence W have to be increased by a factor of 2.
- We tradeoff power for better thermal noise performance.

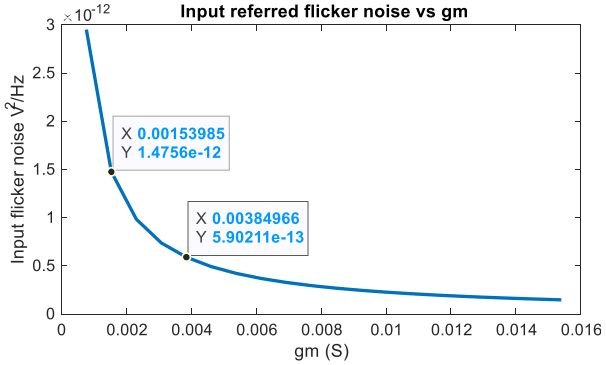
MATLAB session 1

PSD at 1 Hz



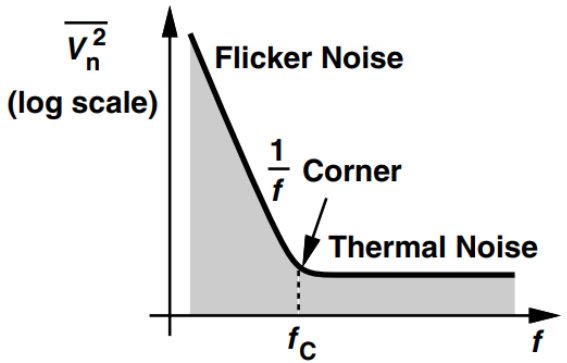




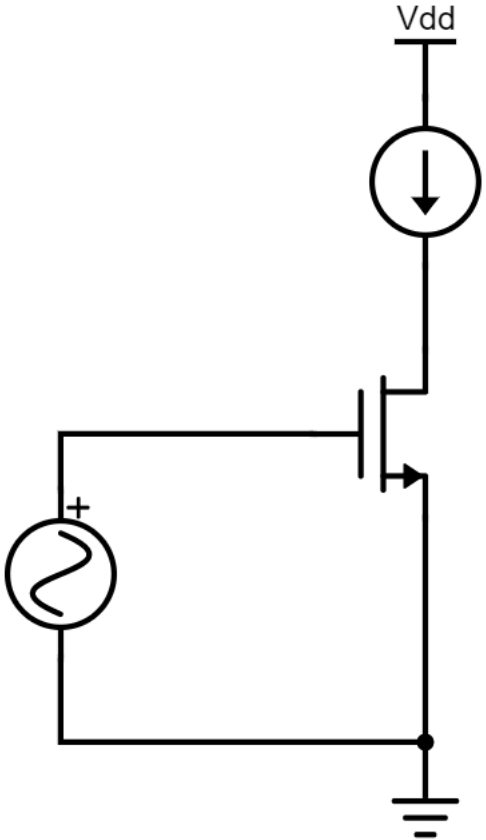


MATLAB session 2

- Speed , noise and Power tradeoff occur quite often in designing circuits.
- Now keep the current constant ids=200u.
- See the frequency where both noise spectral densities are equal? This corner is called flicker noise corner of the device.
- Remember flicker noise spectral density varies as $\frac{C}{f}$ while thermal noise is white.



MATLAB session 3



Fix I_{BIAS}, increment V_{OV} at each step, plot how the width/thermal noise/flicker noise changes with g_m

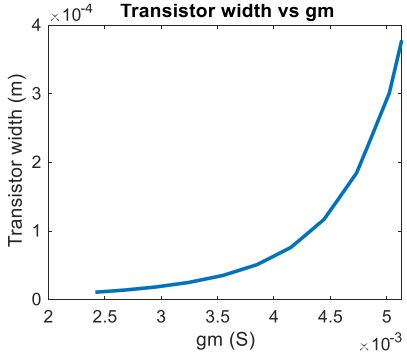
MATLAB session 3

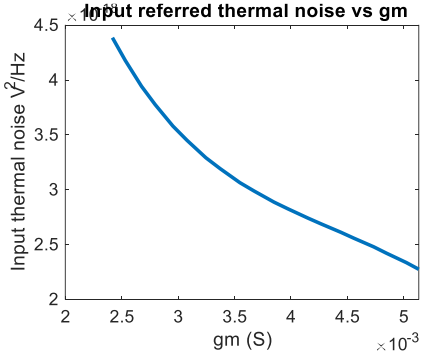
- In this exercise we will keep I_{ds}=200u and vary V_{ov} to change gm.
- gm=2 $\frac{I_{ds}}{V_{ov}}$
- Vary the V_{ov} from 10m to 200m .
- Plot Transistor width vs gm.
- Also plot input referred thermal noise , flicker noise vs gm.
- Keep fnoise_spot1 parameter as 1Hz for now. It is the frequency at which spot noise is obtained.

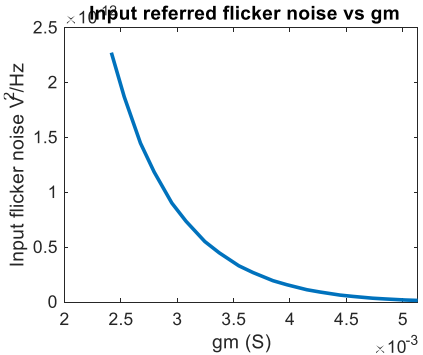
MATLAB session 3

- $I_{ds} = \frac{W}{2L}\mu_n C_{ox} V_{ov}^2$ and hence $\sqrt{\frac{2I_{ds}}{W\mu_n C_{ox}}} = V_{ov}$
- $gm = 2\frac{I_{ds}}{V_{ov}}$ or $gm = \sqrt{\frac{2I_{ds}C_{ox}\mu_n W}{L}}$
- $S_{thermal}(f) = \frac{4kT\gamma}{gm}$ and hence $S_{thermal}(f)\ \alpha\ \frac{1}{\sqrt{W}}$
- $S_{flicker}(f) = \frac{K}{C_{ox}WLf}$ and here f=1Hz .
- To decrease input referred thermal noise by a factor of 2 keeping same I_{ds} , hence W have to be increased by a factor of 4.
- We tradeoff speed for better thermal noise performance.

MATLAB session 3









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